



A review of ethnoveterinary botanical medicines used in Uganda: Their phytochemistry, bioactivity and toxicity

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ABSTRACT

Ethnopharmacological relevance: The use of ethnoveterinary medicine remains a prominent aspect traditional medicine in Uganda. Many rural farmers have and still depend on it for managing their livestock. This review gives a comprehensive over view of ethnoveterinary research in Uganda over a period of quarter a century from 1999 to 2024.

Aim of the study: The aim of this study is to give an over view of the ethnoveterinary medicine in Uganda, entailing the different practices and medicinal plant species used, their bioactivity, phytochemistry and safety/toxicity.

Materials and methods: A literature search was conducted using available databases including ScienceDirect, PubMed, Scopus and Google Scholar. Dissertations, theses, books and technical reports were also searched. The search focused on diseases/ethnoveterinary uses and the animals treated, the method of preparation using the plants, the plant parts used, the bioactivity of the plants in relation to the mentioned traditional uses, their phytochemical composition and toxicity.

Results: The practice of ethnoveterinary medicine is widespread and commonplace in Uganda, especially among the pastoral communities. We recorded the use of 245 plant species from 71 families used against several animal diseases and conditions. Most of the plant species used were from the Fabaceae (38), Asteraceae (16) and Euphorbiaceae (13) families. Most of the plant species were herbs (48 %), followed by shrubs (26 %) and trees (25.4 %) and the most commonly used plant parts used in ethnoveterinary medicine were the leaves (34.2 %), roots (21.0 %) and bark (14.1 %). Of these, 171 (68.9 %) had previous reports supporting one or more of the mentioned traditional uses in different studies *in vitro*. Most of the plant species (207, 84.5 %) had some information on their phytochemical composition known while only 104 (41.9 %), had some information on their safety or toxicity reported in previous studies.

Conclusions: Uganda has a rich repertoire of medicinal plant species used in ethnoveterinary medicine to treat a variety of animal diseases. Most of the plant species used have however not been scientifically evaluated for their efficacy, bioactivity and toxicity. We recommend a systematic study of the medicinal plants species and practices used in ethnoveterinary medicine in Uganda.

1. Introduction

Ethnoveterinary medicine (EVM) is a popular practice among people in low-income nations, incorporating traditional beliefs, knowledge, skills, and practices for managing animal disease (Caudell et al., 2017).

Communities throughout the world use a wide range of EVM to prevent, diagnose, and treat illnesses that affect their livestock and companion animals (Köhler-Rollefson et al., 2008). These practices provide special insights into the interactions that exist between people, animals, and the environment (Batool et al., 2025). Ethnoveterinary medicine has been

Abbreviations: ECF, East Coast Fever; NCD, Newcastle disease; FMD, Foot and Mouth Disease; CNS, Central Nervous System; GLA, Gamma-linolenic acid; THC, Tetrahydrocannabinol; CBD, Cannabidiol; AST, Aspartate Amino Transferase; ALT, Alanine Amino Transferase.

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essential to preserving the health and welfare of many animal species across borders and civilizations. For instance, the Maasai people of East Africa have long relied on the therapeutic qualities of plants like *Vachellia tortilis* (Forssk.) to treat illnesses in their cattle, including pneumonia and diarrhoea (Köhler-Rollefson et al., 2008).

Despite its historical and cultural value, ethnoveterinary medicine is relevant because it presents viable answers to current problems in animal healthcare. There is increased interest in investigating the effectiveness of ethnoveterinary techniques and incorporating them into mainstream veterinary care as traditional veterinary therapies encounter obstacles such as cost, accessibility, and antimicrobial resistance (Mathias, 2004). However, ethnoveterinary medicine is frequently disregarded or ignored in favour of traditional biomedical treatments, despite its potential benefits. This neglect jeopardizes the resilience of communities and ecosystems that depend on traditional veterinary knowledge, in addition to ignoring centuries' worth of accumulated wisdom (McCorkle and Mathias-Mundy, 1992).

Many households around the world depend on their livestock for food, clothing, labour, fertilizer, and money. Animals also serve as a store of wealth and a medium of trade for millions of people around the world (Raikwar and Maurya, 2015). In sub-Saharan Africa, the livestock industry significantly contributes the national economies of various countries (Eiki et al., 2021). In Uganda, rearing livestock accounted for 7.5 % of the country's GDP in 2009, making it a major economic activity (Gradé et al., 2009). The most evident way that livestock might help reduce poverty is by serving as a means of subsistence for the impoverished (Ahuja and Redmond, 2004; Perry and Grace, 2009). Despite its significance as a major contributor to GDP, the livestock industry is plagued by many challenges which limit its development (Eiki et al., 2021). Some of the common challenges include several animal illnesses that lower farm productivity, lack of access to markets and proper medication.

The livestock industry's challenges are not confined to a single country. In Uganda, the public veterinary sector struggles with significant resource shortages, including a lack of adequately trained paraprofessionals and insufficient veterinary services (Nabatanzi et al., 2025). These gaps pose serious risks to animal health. However, paraprofessionals play a crucial role in assisting small-scale farmers who have limited access to public veterinary services (Arvidsson et al., 2022). Research indicates that nearly 90 % of drug retailers and veterinary practitioners lack specialized training in veterinary medicine, which heightens the risk of antibiotic misuse and contributes to antimicrobial resistance (AMR) (Dione et al., 2021). There is also wide spread use of medicinal plants in ethnoveterinary medicine. However, almost all plants are collected from the wild (Gradé et al., 2009), leaving the farmers to the mercies of nature. Farmers in Uganda are also threatened by climate change but have been found to maintain resilience such challenges through local interconnections and agroecology (Le Goff et al., 2022). Challenges of implementing sustainable organic practices in the Ugandan livestock sector threaten its future development. Some of the most pressing challenges include vectors and vector-borne diseases, and limited education, research, and support to organic livestock production (Nalubwama et al., 2011).

The aim of this study was therefore to examine the current state of ethnoveterinary medicine in Uganda, by answering questions about which ethnoveterinary practices and medicinal plant species used in Uganda, what is known about their bioactivity, phytochemistry and safety/toxicity.

2. Materials and methods

Information was retrieved from scientific databases such as Web of Science, ScienceDirect, Google Scholar, Scopus, PubMed and Chemical Abstracts Services. Books, dissertations, theses and technical reports contained from the Makerere University library. Keywords used for the search engines were "Uganda" or "Ugandan" in conjunction with

"ethnoveterinary", "livestock", "traditional animal medicine" and "ethnoveterinary medicine", over the last 25 years from 1999 to 2024. Whereas some of the papers cited used synonyms of some of the plant species, we have used only the accepted names of the plant species.

The inclusion criteria for the research articles selected were: (i) studies focusing on ethnoveterinary botanical medicines in Uganda, (ii) research papers published in peer-reviewed journals (iii) articles with identified phytochemicals and bioactivity studies, (iv) reports on toxicity and safety assessments (v) papers published in English. Studies on human ethnomedicine, articles without botanical authentication, papers lacking scientific validation or phytochemical analysis and non-peer-reviewed sources (e.g., newspaper articles, blogs) were excluded.

To be included in the review, a publication was expected to include details of a specific veterinary use or indication, for example, a plant used for tick control or to expel intestinal parasites in animals. The scientific name of the plant species should have been included as well as the method of preparation and administration should preferably have been recorded. The animal species or "livestock species" was grouped into cattle, goats, sheep, horses, dogs, poultry and donkeys.

The search was done in two phases: The first phase involved identifying research articles reporting the use of plants in ethnoveterinary medicine in Uganda. A total of 245 plant species from 9 studies were identified. This could be due to the fact that only a few studies specifically focusing on ethnoveterinary medicine have been done in Uganda. In the second phase, we systematically searched the literature on the individual plant species for information on their phytochemistry, bioactivity and toxicity. Only research papers that reported bioactivity that was directly relevant to the disease or traditional use were included in the study. For instance, if a plant species was traditionally used for treating Newcastle disease which is a viral disease, we specifically searched for any supporting scientific studies on the antiviral properties of this same plant against Newcastle virus, and if the plant was specifically traditionally used for treating parasitic infestations or wounds, we looked supporting literature on the bioactivity of the said plant species against helminths or bacteria, and so on. However, the information of the phytochemistry and toxicity of the plant species was recorded irrespective of any bioactivity. The scientific names of the plant species in the study were re-checked and named according to the Kew data base <http://powo.science.kew.org> and the plant families were verified using the Angiosperm phylogeny group IV <https://www.gbif.org/dataset/a8ab13c-52ed-4754-b838-aeff74c79718>.

3. Ethnoveterinary medicine use in Uganda

We recorded 245 plant species as being used in ethnoveterinary medicine in Uganda (Table 1). Of these, 170 (69.48 %) had some information reported in the literature supporting one or more of the mentioned traditional uses in different studies, especially *in vitro*. The shortage of *in vivo* studies to validate traditional ethnoveterinary uses undermines claims of efficacy. Most of the plant species (202, 82.4 %) had some information on their phytochemical composition known, even though in most cases, the specific phytochemicals responsible for specific activity were not known. Even though some of the plant species recorded in this review are not native to Uganda, such as *Thymus vulgaris*, they have been introduced into the country and have found their way into the local materia medica. The least available information was on the safety or toxicity of the used plant species. Only 104 (40.9 %), of all the plant species recorded had some information on their safety or toxicity reported in previous studies. However, even this information was scanty and incomplete, therefore not very useful. The information was either available on the crude extracts of the plant species, and seldom on any specific compound(s) from the plant species. Much of the literature on the toxicity also does not specify the exact toxic doses or ranges while other studies do not even report the animal models used. In other instances, reports on toxicity are based on observations in both animals and humans. This makes it difficult to give an objective and

Table 1
Ethnoveterinary Medicinal plants of Ugamda diseases in sub-Saharan Africa.

Scientific Name and (Family)	Diseases/ethnoveterinary use and animals treated	Method of preparation and use	LF	Part used	Bioactivity related to traditional uses in Ugnada	Phytochemical composition	Toxicity
Acanthaceae							
1. <i>Acanthus eminens</i> C.B.Clarke	Coccidiosis (Bukenya-Ziraba and Kamoga, 2007)	Decoction	Sh	L		Alkaloids, phenylpropanoid glycosides, phenylethanoids, flavonoids, lignan glycosides, megastigmane derivatives, triterpenoids, steroids, phenols (Matos et al., 2022)	
2. <i>Crabbea velutina</i> S. Moore.	Compaction (Graté et al., 2009)	Infusion/decoction/used externally as a paste, poultice, or wash (Graté et al., 2009).	T	Fl			
3. <i>Barleria acanthoides</i> Vahl	Fattens donkey (Graté et al., 2009)	Directly given as fodder in donkeys	H	L		Barlerin, acanthoside, shanzhiside, verbascoside, apigenin, luteolin, kaempferol, quercetin, rutin, caffeic acid, chlorogenic acid, and harpagide	Methanolic extract was highly toxic to mice (>2000 mg/kg). Leaf ethanolic extract not toxic in rats at 5000 mg/kg.
Amaranthaceae							
4. <i>Amaranthus dubius</i> Mart. ex Thell.	Helminthosis in livestock (Byaruhanga et al., 2015)	Crush fresh roots, mixed water and the liquid extract is used. Drenching (Byaruhanga et al., 2015)	H	R		Flavonoids and phenolic e.g., vanillic gallic, ellagic acid, flavonoids e.g., catechin, luteolin, and quercetin, phytosterols (Kipkemoi et al., 2023).	
5. <i>Amaranthus hypochondriacus</i> L.	Diarrhoea and weakness in chicken (Zirintunda et al., 2024)	Crushing	H	L			
6. <i>Chenopodium × borbasii</i> Murr	Gastrointestinal disorders, skin infections, parasitic infestations in cattle, goats (Graté et al., 2009), coccidiosis, Fowl typhoid (Bukenya-Ziraba and Kamoga, 2007), measles and diarrhoea in cattle (Tabuti et al., 2003a), wounds in chicken (Zirintunda et al., 2024)	Water extracts, decoction administered orally or topically (Graté et al., 2009). Decoction, plus ash filtrate (Bukenya-Ziraba and Kamoga, 2007), infusion with salt/with leaves <i>Cannabis sativa</i> and <i>Gymnanthemum amygdalinum</i> . Add salt and powder soap (OMO®) to infusion. (3) (Tabuti et al., 2003a)	H	L, Sd	Antimicrobial (Rahman et al., 2023)	Phenols, alkaloids, glycosides, saponins, and flavonoid (Khan et al., 2019).	
7. <i>Achyranthes aspera</i> L.	Rituals	Easy delivery in cows	H	Wp		Saponins, flavonoids, alkaloids, tannins, steroids, triterpenoids, phenolics, betaine, ecdysterone, achyranthine, betavulgarin and rutin (Hossain et al., 2013).	
Amaryllidaceae							
8. <i>Allium cepa</i> L.	Anaplasmosis in poultry and livestock such as cattle, sheep and goats (Graté et al., 2009), Newcastle disease in chicken (Zirintunda et al., 2024)	Water extract with eupet, sediment rock (CaCO ₃) (Graté et al., 2009). Crush bulb in water (Zirintunda et al., 2024)	H	R	Antiviral. Extracts reduces NDV infectivity (Harazem et al., 2019), antimicrobial (Benkeblia, 2004).	Allicin, diallyl disulfide, and quercetin, flavonoids, phenolic acids, saponins, sterols	Toxic threshold varies with animal species, body weight, and type of onion, between 15 and 30 g/kg of onion (Ganesan et al., 2008).
9. <i>Allium sativum</i> L.	Newcastle disease, infectious bronchitis, Bacillary white diarrhoea (Bukenya-Ziraba and Kamoga, 2007), Newcastle disease in chicken (Zirintunda et al., 2024)	Cold water extract/Decoction with different herbs (Bukenya-Ziraba and Kamoga, 2007; Zirintunda et al., 2024), Crush bulb in water (Zirintunda et al., 2024)	H	Bib	Antiviral effects against a velogenic strain of Newcastle Virus Disease (NDV) <i>in ovo</i> accompanied by a reduction in virus infectivity and titer (Doostmohammadian et al., 2020), antimicrobial (Benkeblia, 2004).	Allicin, E-Ajoene, Diallyl trisulfide, Alliin, Allyl methyl sulfide, 2, Diallyl disulfide, Quercetin, allyl mercaptan, Allyl propyl disulfide, thiaceomonone, allyl methyl trisulfide (Okoro et al., 2023).	Safe (Lawal et al., 2016)
Anacardiaceae							

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Table 1 (continued)

Scientific Name and (Family)	Diseases/ethnoveterinary use and animals treated	Method of preparation and use	LF	Part used	Bioactivity related to traditional uses in Ugnada	Phytochemical composition	Toxicity
10. <i>Lannea humilis</i> (Oliv.) Engl.	Calf diarrhoea, switches in cattle, sheep, goats, carmels (Gradé et al., 2009)	Water extract applied orally (Gradé et al., 2009)	T	B, L	Antimicrobial against <i>K. pneumoniae</i> , <i>E. coli</i> , <i>S. pyogenes</i> , <i>S. aureus</i> , <i>Bacillus subtilis</i> , <i>Proteus mirabilis</i> , and <i>Salmonella typhi</i> (Achika et al., 2020)	Phenolics, flavonoids e.g., quercetin, catechin, and epicatechin flavan-3-ol derivatives, tannins and triterpenoids (Malú et al., 2024)	
11. <i>Mangifera indica</i> L.	Newcastle disease	Decoction (Bukunya-Ziraba and Kamoga, 2007)	T	B	Antiviral effects against symptoms of Newcastle disease (Boakye et al., 2024)	Protocatechic acid, catechin, mangiferin, alanine, glycine, γ -aminobutyric acid (Scartezzini and Speroni, 2000), triterpenoid and flavonoids (Scartezzini and Speroni, 2000; Singh et al., 2015)	Leaf extracts are non-toxic in ICR mice in acute and long-term toxicity studies up to 18.4 g/kg bw (Zhang et al., 2014)
12. <i>Ozoroa insignis</i> Delile	Diarrhoea, painful red yes	Water extract	T	B, R	Antibacterial, antifungal, anti-inflammatory (Dube et al., 2021).	Flavonoids, tannins, saponins, terpenoids, and phenolsic.	
13. <i>Searsia natalensis</i> (Bernh. ex C.Krauss) F.A. Barkley	Retained placenta	Fresh fruits and leaves crushed with water, for drenching (Nabukunya et al., 2014).	H	L, Fr			
14. <i>Searsia pyroides</i> (Burch.) Moffett	Diarrhoea, fever, lumpy skin disease	Water extract/fodder	T	R	Anti-bacterial (Mutuku et al., 2020).	Glycosides, alkaloids, flavonoids, terpenoids, tannins, and saponins (Mutuku et al., 2020).	Non cytotoxic at 100 μ g/ml (Mutuku et al., 2020)
Annonaceae							
15. <i>Annona muricata</i> L.	Bacterial infections in chicken (Zirintunda et al., 2024)	Crushing	T	L and F	Antimicrobial, anti-inflammatory, anti-protozoan (Coria-Téllez et al., 2018)	Alkaloids, phenols and acetogenins (Coria-Téllez et al., 2018)	
Apocynaceae							
16. <i>Carissa spinarum</i> L	Anaplasmosis, contagious bovine/pleuropneumonia, chicken pox, diarrhoea, East Coast fever (ECF), general body pains, heartwater, lumpy skin disease, rinderpest, skin disease with intestinal adhesions in cattle, poultry, goats and sheep (Gradé et al., 2009; Tabuti et al., 2003a).	Water extract which is orally applied/infusion with <i>Zanthoxylum chalybeum</i> roots, (Gradé et al., 2009), (1) Mix in warm banana beer and give calf. (2) Infusion prepared after adding <i>Vachellia seyal</i> roots (Tabuti et al., 2003a).	Sh	R	Anti-inflammatory, analgesic, antipyretic, antibacterial, antimalarial, antihelminthic, anticancer, antioxidant, wound healing (Sharma et al., 2023).	Carissone, carindone, carinol, carissic acid, lupeol, β -sitosterol, ursolic acid, oleanolic acid, and quercetin (Asfaw et al., 2022).	Low toxicity in mice and rats (Smyth and Sheridan, 2022).
17. <i>Cascabela thevetia</i> (L.) Lippold	Cough in chicken (Zirintunda et al., 2024)	Crushing (Zirintunda et al., 2024)	Sh	L		Flavonoids mainly rutin and phenolic acids mainly rosmarinic acid (El-Sawi et al., 2020), cardiac glycosides e.g., thevetin 7 peruvoside (Misra et al., 2009).	
18. <i>Orbea dummeri</i> (N.E.Br.) Bruyns	Wounds (Gradé et al., 2009).	Juice spread topically on the animal (Gradé et al., 2009).	H	Fl	Anti-microbial against <i>S. aureus</i> , <i>E. coli</i> , <i>P. aeruginosa</i> and <i>C. albicans</i> .	Pregnane glycosides, flavonoids, and phenolic acids (Adnan et al., 2014).	
19. <i>Desmidorchis retrospiciens</i> Ehrenb.	Helminthosis, ticks in livestock (Byaruhanga et al., 2015)	Fresh roots are crushed, mixed with water and liquid extract used for drenching. Fresh leaves and stem crushed and liquid used for washing the animals (Byaruhanga et al., 2015).	Sh	R, L			
20. <i>Saba comorensis</i> (Bojer ex A.DC.) Pichon	Back pain	Juice and fruit flesh	T	Fr			

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Table 1 (continued)

Scientific Name and (Family)	Diseases/ethnoveterinary use and animals treated	Method of preparation and use	LF	Part used	Bioactivity related to traditional uses in Ugnada	Phytochemical composition	Toxicity
21. <i>Stephanotis rubicunda</i> (K. Schum.) S.Reuss, Liede and Meve	Sore mouth in kids	Gently crush leaves and apply orally	T	Lv		Lupeol, sitosterol, stigmasterol, gallic acid, caffeic acid, quercetin, and kaempferol.	
22. <i>Secamone africana</i> (Oliv.) Bullock	Gastro-intestinal parasites in livestock (Nalule et al., 2013)	Maceration (Nalule et al., 2013)	Sh	Wp	Anthelmintic (Nalule et al., 2013)	Tannins, alkaloids, saponins, flavonoids, steroids, glycosides, triterpenes, coumarin derivatives, anthocyanocides, anthracenocides (Nalule et al., 2013)	
Araceae							
23. <i>Pistia stratiotes</i> L.	ECF in cattle (Tabuti et al., 2003a)	Infusion made after adding roots or leaves of <i>Oncoba spinosa</i> (Tabuti et al., 2003a).	H	Wh		Alkaloids, flavonoids, glycosides and phlobatanins (Lourenção et al., 2021; Ma et al., 2019)	
Asparagaceae							
24. <i>Agave sisalana</i> Perrine	Fowl typhoid (Bukenya-Ziraba and Kamoga, 2007), Newcastle disease in chicken (Zirintunda et al., 2024)	Warm to extract juice (Bukenya-Ziraba and Kamoga, 2007), crush leaves in water (Zirintunda et al., 2024)	Sh	L	Anti-inflammatory activity (Costa et al., 2023).	Steroidal saponins, dongnosides C-E (Yi et al., 1989), kaempferol (Costa et al., 2023).	The hydrolyzed extract was not toxic in rats orally in a single dose, up to 1000 mg/kg <i>in vivo</i> (Costa et al., 2023).
25. <i>Asparagus racemosus</i> Willd.	Induce follicle growth and estrus in cows (Sarswat and Purohit, 2020), genital diseases, hydrocoel, swollen testis, anaplasmosis (Gradé et al., 2009), ECF in cattle (Tabuti et al., 2003a)	Water extract, oral and topical administration (Gradé et al., 2009), infusion with a little salt (Tabuti et al., 2003a).	H	R		Steroidal saponins, flavonoids, phenolics, tannins, alkaloids, and terpenoids, asparagusic acid, asparanin A, asparoside A, rutin, quercetin, kaempferol, and protodioscin (Alok et al., 2013).	Delayed effects of toxicity in rodents via relative organ weights of the liver, spleen, kidney and heart at a 700 mg/kg and damage to vital organs (Soren and Yadav, 2022)
26. <i>Dracaena dawei</i> (Stapf) Byng & Christenh	Singamus (worms) (Bukenya-Ziraba and Kamoga, 2007)	Cold water extract (Bukenya-Ziraba and Kamoga, 2007)	H	L			
27. <i>Dracaena nilotica</i> (Baker) Byng & Christenh.	Coccidiosis (Bukenya-Ziraba and Kamoga, 2007)	Cold water extract (Bukenya-Ziraba and Kamoga, 2007)	Sh	L			
Asteraceae							
28. <i>Acmella caulirhiza</i> Delile	Tooth and stomach issues in animals (Bussmann et al., 2020a)	Crush and mix with water and administering orally	H	L, R	Antioxidant, antiinflammatory, analgesic, antibacterial and antifungal against <i>S. aureus</i> , <i>E. coli</i> , <i>P. aeruginosa</i> and <i>C. albicans</i> (Noubissi et al., 2021; Sharma et al., 2022), antiinflammatory and anticancer (Tienoue Fotso et al., 2024).	Flavonoids, phenolics, tannins, saponins, alkaloids and terpenoids (quercetin, kaempferol, caffeic acid, chlorogenic acid and spilanthal) (Tienoue et al., 2023).	Could have deleterious effects at 500 mg/kg Bw, causing an increase in the relative weight of the brain, uterus, ovaries and a change in some haematological parameters (Tienoue et al., 2023).
29. <i>Ageratum conyzoides</i> L.	Fowl typhoid	Decoction (Bukenya-Ziraba and Kamoga, 2007)	H	L	Antiinflammatory in rats (Moura et al., 2005), the flavonoids showed antiprotozoal activity (Nour et al., 2010)	Alkaloids, flavonoids, terpenes, chromenes, and sterols (Yadav et al., 2019), chromenes, benzofurans and terpenoids (Okunade, 2002)	No apparent hepatotoxicity in rats at 500 mg/kg Bw (Moura et al., 2005). No mutagenicity, clastogenicity/genotoxicity (Palmer et al., 2019)
30. <i>Aspilula africana</i> (Pers.) C.D. Adams	Influence growth, carcass yield, egg and meat quality, anaemia, gastrointestinal tract infections, wound healing, stimulate lactation (Gradé et al., 2009), cattle helminthosis (Matovu et al., 2020)	Decoction, infusions administered orally/topically (Gradé et al., 2009), maceration, drench 0.5L to adult animals and 300 ml to calves (Matovu et al., 2020)	H	L	Antibacterial, Antioxidant, boosts immunity, enhance metabolism (Etim et al., 2021).	Saponins, tannins, alkaloids, flavonoids, terpenoids, phenols, (Adaekwu et al., 2024).	Low toxicity in mice. Acute toxicity and lethality (LD ₅₀ = 6.6 mg/kg Bw) (Taziebou et al., 2007).
31. <i>Aspilula mossambicensis</i> (Oliv.) Wild	Anaplasmosis, East coast fever (Gradé et al., 2009)	Water extract with Solanum spp. Fruits given topically in sheep, goats, cattle and dogs	H	Fl, R	Anti-inflammatory and antiparasitic against nematodes,	Diterpenes e.g., daurenoic and grandifloreneic acids (McMillen,	Large amounts/long term use can harm, livestock. Thiarrubines, can cause liver

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Table 1 (continued)

Scientific Name and (Family)	Diseases/ethnoveterinary use and animals treated	Method of preparation and use	LF	Part used	Bioactivity related to traditional uses in Ugnada	Phytochemical composition	Toxicity
32. <i>Bidens pilosa</i> L.	Eyes painful red rats, mice, dogs, and chicken (Gradé et al., 2009), Newcastle disease and Ectoparasites in chicken (Zirintunda et al., 2024)	Paste in eye (Gradé et al., 2009), crush leaves in water (Zirintunda et al., 2024)	H	R	protozoa, and ticks (McMillen, 2012). Antimicrobial, antiparasitic, anti-inflammatory, analgesic (Woldeamanuel et al., 2022).	2012), thiarubridines, sulfur compounds (Norton et al., 1993). Quercetin, kaempferol, caffeic acid, chlorogenic acid, luteolin, centaureidin, okanin, and aurantiamide (Idris et al., 2023).	damage, bleeding, anaemia, and death (Norton et al., 1993). Well-tolerated (Idris et al., 2023).
33. <i>Gutenbergia cordifolia</i> Benth. ex Oliv.	Pneumonia in sheep and goats (Gradé et al., 2009)	Directly as fodder (Gradé et al., 2009)	H	L	Wound healing, analgesic, antidiarrhoeal, antifungal, anti-inflammatory, antipyretic, anthelmintic	Tannins, saponins, flavonoids, glycosides, phenols, steroids, and alkaloids, cordifolin, cordifolone, cordifolioside, cordifolide etc	
34. <i>Gnaphalium purpureum</i> L.	Eyes painful red	Paste in eye	H	L	Anti-inflammatory, analgesic, antipyretic, antispasmodic, antifungal, antibacterial, and antiviral (Bruschi et al., 2017)	Flavonoids (e.g., luteolin, kaempferol, apigenin, and quercetin), terpenoids, phenolic acids (Bruschi et al., 2017).	
35. <i>Kleinia odora</i> (Forssk.) DC.	Strengthen sick calf	Water extract which is orally applied	H	St	Anti-oxidant (Awuchi and Morya, 2023)	Glycosides, alkaloids, tannins, flavonoids, and terpenoids (Awuchi and Morya, 2023).	
36. <i>Schkuhria pinnata</i> (Lam.) Kuntze ex Thell.	Body weakness in chicken (Zirintunda et al., 2024)	Pounding (Zirintunda et al., 2024)	H	Wp		Flavonoids e.g., pectolinarigenin, eudesmanolide (Kimani et al., 2018), sesquiterpene lactones e.g., schkuhriopinnatolides (Ganzer and Jakupovic, 1990)	
37. <i>Sphaeranthus suaveolens</i> (Forssk.) DC.	Fleas and lice	Water extract, topically	H	Wp	Anti-pest and antiinfective, insecticidal (Laizer and Mbwapo, 2022).	Phenols, terpenoids, steroids, flavonoids, and saponins (Laizer and Mbwapo, 2022).	Safe in rats (Sudi et al., 2021).
38. <i>Sphaeranthus ukambensis</i> Vatke & O.Huffm.	Bloat	Water extract	H	Wp		Carvotanacetone derivatives (1–6) (Pouny et al., 2011).	
39. <i>Tithonia diversifolia</i> (Hemsl.) A.Grey	Fowl typhoid	Decoction (Bukonya-Ziraba and Kamoga, 2007)	H	L	Antibacterial (Dlamini et al., 2020).	Alkaloids, tannins, flavonoids (e.g., luteolin and nepetin), saponins, terpenoids e.g., diversifolide, phenols (Ambrósio et al., 2008; Kuo and Lin, 1999; Olayinka et al., 2015), essential oils with α - and β - pinene, β -caryophyllene, isocaryophyllene, nerolidol, α -copaene, α -gurjunene, cyclodecene (Agboola et al., 2016; Moronkola et al., 2007; Wanzala et al., 2016). sesquiterpene lactones (Passoni et al., 2013).	The ethanol extract is relatively safe at ≥ 100 mg/kg. Prolonged use of chlorogenic acid derivatives and sesquiterpene lactones damages liver and kidneys (Passoni et al., 2013), with both haematological and acute toxic effects on the kidney and liver (LD ₅₀ > 1600 mg/kg/day) in rats
40. <i>Tagetes minuta</i> L.	Lumpy skin disease, lice, poultry respiratory infection (Gradé et al., 2009)	Water extract poultice (Gradé et al., 2009)	H	Wp	Antibacterial and antioxidant (Igwaran et al., 2017).	Monoterpenes and sesquiterpenes, e. g., tagetone, dihydrotagetone, ocimenone, limonene, β -caryophyllene and zingiberene (Igwaran et al., 2017).	Considered safe, but its oil & extracts can cause photo-toxicity (Coenraads, 2016)
41. <i>Gymnanthemum amygdalinum</i> (Delile) Sch.Bip.	Parasitic infections, diarrhoea, wound healing and inflammatory conditions in cattle, goats and sheep (Gradé et al., 2009), Newcastle disease, coccidiosis, Fowl typhoid (Bukonya-Ziraba and Kamoga, 2007), Foot and Mouth Disease (FMD), Fever, cough and diarrhoea in cattle (	Water extracts applied orally/ topically (Gradé et al., 2009), decoction (Bukonya-Ziraba and Kamoga, 2007). Maceration/decoction and drench (Matovu et al., 2020), infusion with leaves of <i>C. opulifolium</i> & <i>S. occidentalis</i> (Tabuti et al., 2003a).	Sh	L	Antihelmintic (Degu et al., 2024), antibacterial, antifungal, antiparasite, antihelmintic, antiviral (Wang et al., 2018)	Flavonoids, alkaloids, saponins, tannins, triterpenoids, sesquiterpene lactones, steroids, cardiac and cyanogenic glycosides, oxalates, phytates, and phenols (Adiukwu et al., 2011).	

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Table 1 (continued)

Scientific Name and (Family)	Diseases/ethnoveterinary use and animals treated	Method of preparation and use	LF	Part used	Bioactivity related to traditional uses in Ugnada	Phytochemical composition	Toxicity
42. <i>Tridax procumbens</i> L.	Matovu et al., 2020; Tabuti et al., 2003a) Cough and diarrhoea in chicken (Zirintunda et al., 2024)	Crushing	H	L	Anti-inflammatory and antimicrobial (Gubbiveeranna and Nagaraju, 2016)	Procumbenase (Gubbiveeranna et al., 2024), octadecanoic acid, icaricide, loliolide, neoxanthin, antheraxanthene (Abubakar and Haque, 2020). Has phytotoxic compounds that are probably flavonoids (Mecina et al., 2016).	No clinical signs of toxicity in BALB/c mice but pure oxylipin isolated from <i>T. procumbens</i> showed signs of toxicity and death in mice (Burgos-Pino et al., 2023).
Apocynaceae							
43. <i>Adenium obesum</i> (Forssk.) Roem. & Schult.	Mange (Gradé et al., 2009), ticks in livestock (Byaruhanga et al., 2015)	Poultice (Gradé et al., 2009), a piece of fresh root tuber is cut, crushed, and mixed with water to obtain a solution which is used. Wash animal body with the solution (Byaruhanga et al., 2015).	Sh	B	Antibacterial, antiinflammatory, analgesic (Suleiman and Brima, 2021).	Cardiac glycosides, flavonoids, terpenoids, pregnanes, and alkaloids, oleadriin, neridienone, adenin, adenobetin (Sornalakshmi et al., 2021).	Highly toxic. Can cause cardiac arrhythmias, respiratory depression, liver and kidney impairment, and death when ingested in high doses/over a long period of time (Muyiwa et al., 2019).
44. <i>Gomphocarpus fruticosus</i> (L.) W. T.Aito	Back pain Diarrhoea (Gradé et al., 2009)	Water extract which is orally applied (Gradé et al., 2009)	H	R	Hepatoprotective, wound healing, anti-inflammatory, analgesic, antipyretic, antidiarrheal, antispasmodic, antifungal, antibacterial, & antiviral (Mphahlele and Viljoen, 2026).	Lupeol, ursolic acid, caffeic acid, chlorogenic acid, luteolin, apigenin, kaempferol, ferulic acid, and gomphoside, coumarin glycoside gomphoflorosides and gomphofruticosine (Dushimemariya, 2019; Mphahlele and Viljoen, 2026)	Toxic nature to livestock & humans, and has caused deaths in cattle, sheep & poultry (Van Wyk et al., 1997).
45. <i>Mondia whitei</i> (Hook.f.) Skeels	Body weakness in chicken (Zirintunda et al., 2024)	Pounding	H	R		2-hydroxy-4- methoxybenzaldehyde phenolics, flavonoids and tannins (Aremu et al., 2011; Bouba et al., 2010).	Aqueous extracts are non-toxic in acute and sub- chronic toxicity on Wistar rats (Joseph et al., 2015).
46. <i>Cynanchum viminale</i> (L.) L.	Bloat (Gradé et al., 2009)	Water extract (Gradé et al., 2009)	H	Wp		Alkaloids, flavonoids, phenols, saponins, steroids, and terpenoids (Arora and Meena, 2018), sarcovimside A-C, pregnane glycosides (Vleggaar et al., 1993)	Toxic to sheep (Jackson and Shone, 1958), Poisoning from <i>S. viminale</i> resulted in beak patting, muscular tremors and head flopping, and death in ostriches (Cooper, 2007).
Araliaceae							
47. <i>Steganotaenia araliacea</i> Hochst.	Lumpy skin disease, snake bite, snake spit in cattle, sheep, goats, chicken and pigs (Gradé et al., 2009) coccidiosis (Bukenya-Ziraba and Kamoga, 2007), ECF (Tabuti et al., 2003a)	Water extract, decoction (Gradé et al., 2009; Bukenya-Ziraba and Kamoga, 2007), infusion (Tabuti et al., 2003a)	T	B, L, R	Antibacterial, antimalarial, antidiabetic, antileukemic, and antitubulin (Capistrano et al., 2015).	Alkaloids, flavonoids, terpenoids, saponins, tannins, and phenolics, steganacin, steganangin, steganolide A, steganol, steganone, episteganangin, steganoate A and B, cumene, xylene, citronellol, cadinanol, arcurcumene, and α -guaiene (Capistrano et al., 2015).	Dichloromethane extracts were, generally more toxic than methanol and ethanol extracts to Vero cells ≥ 1 μ g/ml cytotoxic (Beuscher et al., 1994).
Aristolochiaceae							
48. <i>Aristolochia littoralis</i> Parodi	East coast fever (ECF) in cattle (Tabuti et al., 2003a)	Infusion (Tabuti et al., 2003a).	H	L	Abortive activity of the genus in mice (Pakrashi and Shaha, 1977)	Alkaloids, steroid, tannins (NK and MP, 2022), aristolochic acids 1 (Wu et al., 2001).	Aristolochic acids (Wu et al., 2001) responsible for the nephrotoxic and genotoxic effects (Kumar et al., 2003)
Asphodelaceae							
49. <i>Aloe dawei</i> A. Berger	Anaplasmosis, fly repellent, skin infections, pneumonia, in sheep cattle and goats bloody calf diarrhoea, trypanosomiasis and retained placenta (Gradé et al., 2009).	Water extract, administered orally Poultice applied topically (Jambwa and Nyahangare, 2020).	H	L, Ex	Anti-inflammatory and wound-healing (Jambwa and Nyahangare, 2020).	Polysaccharides, anthraquinones, and flavonoids (Jambwa and Nyahangare, 2020).	Safe for animals but excessive amounts may cause vomiting, diarrhoea, dehydration, and liver impairment (Jambwa and Nyahangare, 2020).
50. <i>Aloe</i> sp	ECF, infections, wounds, skin diseases, gastrointestinal disorders,	Water extract given orally	H	L	Anti-inflammatory, antimicrobial, antiparasitic and	Anthraquinones, polysaccharides, phenolics (Aremu and Lawal, 2022)	Aloe spp are generally safe for animals, but can cause diarrhoea, vomiting,

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Table 1 (continued)

Scientific Name and (Family)	Diseases/ethnoveterinary use and animals treated	Method of preparation and use	LF	Part used	Bioactivity related to traditional uses in Ugnada	Phytochemical composition	Toxicity
51. <i>Aloe secundiflora</i> var. <i>tweedieae</i> (Christian) Wabuye	and internal parasites in livestock and poultry (Gradé et al., 2009). Anaplasmosis (Gradé et al., 2009).	H	H	Ex	wound-healing properties (Aremu and Lawal, 2022)		electrolyte imbalances and loss of appetite (Aremu and Lawal, 2022).
52. <i>Aloe vera</i> (L.) Burm.f.	Helminthosis in all livestock and poultry (Nabukenya et al., 2014), coccidiosis, NCD, cough and fowl cholera, histomoniasis, infectious bronchitis in chicken, pullorum/salmonellosis in chicken (Zirintunda et al., 2024)	Slice and boil the leaves and give to drink/add to feeds (Nabukenya et al., 2014), crushing (Zirintunda et al., 2024)	H	L	Antiseptic activity, antifungal, antibacterial and antiviral (Shariff Moghaddasi and Res, 2011)	Salicylic acid, prostaglandin precursors, lignins, saponins, sterols (Atherton, 1998), anthraquinone glycosides: aloin, aloe-emodin, barbaloin (Shariff Moghaddasi and Res, 2011), lupeol, cinnamonic acid, phenols (Shariff Moghaddasi and Res, 2011)	Reproductive toxicity in Sprague Dawley rats (Shah et al., 1989), carcinogenic activity in F344/N rats (National Program Toxicology, 2015)
53. <i>Aloe peglerae</i> Schönland	Chronic respiratory disease and pullorum/salmonellosis in chicken (Zirintunda et al., 2024)	Crushing	H	L			
Asparagaceae							
54. <i>Chlorophytum comosum</i> (Thunb.) Jacques	Lactation in cattle (Tabuti et al., 2003a)	Infusion. Add salt.	H	B			
55. <i>Dracaena fragrans</i> (L.) Ker Gawl.	Diarrhoea, brucellosis (Gradé et al., 2009).	Water extract which is orally applied. Boiling and drenching all cows in the morning (Gradé et al., 2009),	Sh	R			
Bignoniaceae							
56. <i>Kigelia africana</i> (Lam.) Benth.	Bloody calf diarrhoea, contagious, bovine/caprine pleuropneumonia, eye infection, heartwater, pink eye (Gradé et al., 2009), coccidiosis and NCD (Zirintunda et al., 2024)	Water extract and applied orally, paste from ashes, Paste with leaf ash and animal butter, in the eye (Gradé et al., 2009), pound fruit (Zirintunda et al., 2024)	T	L, B, R	Anti-inflammatory, antibacterial, antifungal, antiviral, antihelminthic, and antiprotozoal (Nabatanzi et al., 2020).	Triterpenoids, flavonoids, phenylpropanoids, euscaphic acid (Bussmann et al., 2021), iridoids, isocoumarins, naphthoquinones, phenylpropanoids, furanone derivatives, flavonoids and sterols (Osman et al., 2017)	
Brassicaceae							
57. <i>Brassica oleracea</i> L.	Diarrhoea in chicken (Zirintunda et al., 2024)	Crushing	H	L	Anti diarrheal activity (Shakar et al., 2012).	Alkaloids, flavonoids, glucos-inolates, steroids, terpenoids, benzenoids (Kabouw et al., 2010; Nielsen et al., 1993; Park et al., 2014; Thangam et al., 2013)	Sedation, lethargy and drowsiness in sprague dawley rats at 2000 mg/kg and 4000 mg/kg (Shah et al., 2016).
Bromeliaceae							
58. <i>Ananas comosus</i> (L.) Merr.	ECF in cattle (Tabuti et al., 2003a), cough in chicken (Zirintunda et al., 2024)	Crushing, Infusion (Tabuti et al., 2003a)	H	Fr, L			
Burseraceae							
59. <i>Commiphora africana</i> (A. Rich) Engl.	Fly and maggot control, fresh wounds (Gradé et al., 2009), ticks in livestock (Byaruhanga et al., 2015)	Exudate and powdered sap and topically applied (Gradé et al., 2009), Fresh roots crushed, mixed with water, and used for washing the animals (Byaruhanga et al., 2015)	T	L ex.	Analgesic, antimalarial, antibacterial, and anthelmintic actions (Paraskeva, 2008).	Alkaloids, flavonoids, coumarins, and phenolics (Okwute et al., 2022).	
60. <i>Commiphora kua</i> (R.Br. ex Royle) Vollesen	Eyes painful red (Gradé et al., 2009)	Exudate Powdered sap, in eye (Gradé et al., 2009)	T	Ex		B-elemene, α -selinene, cadina-1,4-diene, germacrene B, α -copaene, t-muuirolol, caryophyllene oxide, and α -cadinol (Ali et al., 2009)	

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Scientific Name and (Family)	Diseases/ethnoveterinary use and animals treated	Method of preparation and use	LF	Part used	Bioactivity related to traditional uses in Ugnada	Phytochemical composition	Toxicity
Cactaceae							
61. <i>Opuntia cochenillifera</i> (L.) Mill.	Compaction, low milk production, retained placenta (Gradé et al., 2009)	Water extract/decoction (Gradé et al., 2009)	H	Fr, St		Alkaloids, phenolic compounds, terpenoids, flavonoids, and saponins (da Cruz Filho et al., 2019).	
Cannabaceae							
62. <i>Cannabis sativa</i> L.	Heartwater, pneumonia, diarrhoea, trypanosomiasis, ECF (Gradé et al., 2009), coccidiosis, worms, infectious bronchitis, bacillary white diarrhoea, FMD, fever, cough in cattle and chicken and fowl typhoid (Matovu et al., 2020; Tabuti et al., 2003a), NCD (Bukonya-Ziraba and Kamoga, 2007; Zirintunda et al., 2024).	Vaporizing/smoking dried buds, drinking/eating extracts, and taking capsules. Crush fresh leaves/add dried powder and mix with water (Bukonya-Ziraba and Kamoga, 2007),/maceration, decoction and drenching (Matovu et al., 2020), decoction/infusion with <i>S. longipedunculata</i> roots.	H	L, R, Sd	Antibacterial (Shah et al., 2020), analgesic (Bonini et al., 2018).	Cannabinoids, e.g., tetrahydrocannabinol (THC) and cannabidiol (CBD)(Malú et al., 2024), terpenophenolic compounds (cannabinoids), terpenoids, flavonoids, alkaloids, nitrogen compounds (Bonini et al., 2018).	
Canellaceae							
63. <i>Warburgia ugandensis</i> Sprague	Cough and diarrhoea in chicken (Zirintunda et al., 2024)	Water extract (Zirintunda et al., 2024)	T	B	Antimicrobial activity (Olila and Opuda-Asibo, 2001)	Drimane sesquiterpenes e.g., ugandensidial, polygodial, warburganal and isopolygodial (Were et al., 2015)	Highly cytotoxic to U87 cells (Anywar, 2020; Anywar et al., 2022), cytotoxic to intestinal epithelial cells IEC-6, with IC ₅₀ values < 50 µg/ml (Mwitari et al., 2013).
Capparaceae							
64. <i>Capparis fascicularis</i> DC. var.	Heartwater, Lumpy skin disease, snakebite (Gradé et al., 2009).	Water extract with <i>Carissa edulis</i> and <i>S. araliacea</i> leaves and root, topically given (Gradé et al., 2009).	Sh	B, R			
65. <i>Capparis tomentosa</i> Lam.	Heartwater, anaplasmosis (Gradé et al., 2009).	Water extract	Sh	B, R	Anti-inflammatory, analgesic, antipyretic, antimicrobial (Tekulu et al., 2020)	Cappariloides, tomentosin, tomentocardin, and tomentodione (Gebrehiwot and Chaithanya, 2020; Tlili et al., 2011).	.
66. <i>Capparis</i> spp.	Anaplasmosis, Black quarter, contagious bovine/caprine pleuropneumonia, ECF, swollen lymph nodes, pain killer, sudden death, fever (Gradé et al., 2009).		Sh	B, R			
67. <i>Maerua edulis</i> (Gilg & Gilg-Ben.) DeWolf	Abscess, dysentery, intestinal worms in goats, cattle, sheep(Gradé et al., 2009).	Water extract	H	R		Flavonoids, alkaloids, saponins, tannins, and lignans (Lagu et al., 2025).	
68. <i>Maerua gilgii</i> Schinz	Ticks in livestock (Byaruhanga et al., 2015)	Crushed with water use for washing animal (Byaruhanga et al., 2015).	H	R			
69. <i>Maerua parvifolia</i> Pax	Acute eye syndrome, painful red eyes, pinkeye in sheep, goats, cattle (Gradé et al., 2009).	Paste in eye/masticate leaves and spit in eye/decoction (Gradé et al., 2009).	Sh	L, R	Antibacterial s, antipyretic, and anthelmintic (Abdalla and McGaw, 2020).	Alkaloids, flavonoids, glycosides (Abdalla and McGaw, 2020).	
Caricaceae							
70. <i>Carica papaya</i> L.	Intestinal parasites in livestock (Gradé et al., 2009). Retained placenta (Matovu et al., 2020), coccidiosis, fowl typhoid in chicken (Zirintunda et al., 2024)	Seeds fed to animals (Gradé et al., 2009), chop young fruits/grind 100g of seed, pound and mix with feed/put in the mouth (Matovu et al., 2020)/pound leaves, fruits, seeds, flowers leaves and administer (Zirintunda et al., 2024).	T	Sd	Antibacterial and anti-inflammatory (Lim et al., 2021).	Carpaine, pseudocarpaine, benzyl isothiocyanate, benzyl glucosinolate, papain, chymopapain, and lycopene (Singh et al., 2020).	Safe and well-tolerated use in humans and animals (Lim et al., 2021).

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Caryophyllaceae							
71. <i>Drymaria cordata</i> (L.) Willd. ex Schult.	Coccidiosis (Bukenya-Ziraba and Kamoga, 2007)	Cold water extract	H	Wp	Antimicrobial activity (Antonyan et al., 2016)	Alkaloids, flavonoids, phenols, tannins and saponins (Bhattacharyya et al., 2019)	Toxic in zebra fish-visible developmental defects, e.g., heartbeat rate, less pigmentation, oedema, spinal curvature, immature yolk sac and reduced hatching rate and a slow growth (Patra et al., 2023).
Celastraceae							
72. <i>Elaeodendron buchananii</i> (Loes.) Loes.	Trypanosomiasis in cattle, sheep, goats, and camels, stimulant (Graté et al., 2009)	Decoction (Graté et al., 2009)	T	L	Antiplasmodial, antibacterial & anti-inflammatory properties (Jennifer et al., 2022).	Flavonoids, terpenoids, alkaloids, and phenolics e.g., quercetin, kaempferol, lupeol, elaeodendrine and elaeodendroside (Odak et al., 2018).	
73. <i>Gymnosporia senegalensis</i> (Lam.) Loes.	ECF in cattle (Tabuti et al., 2003a)	Infusion; 125 ml given to calf twice a day (Tabuti et al., 2003a)	Sh	R	Antimicrobial activity (Makgatho et al., 2018)	Maytenoic acid, pristimerin, lupenone, β -amyryn and β -sitosterol (Makgatho et al., 2018)	
Cleomaceae							
74. <i>Cleome gynandra</i> L	Trachoma (Graté et al., 2009), cough in chicken (Zirintunda et al., 2024)	Water extract which is orally applied (Graté et al., 2009), pound (Zirintunda et al., 2024)	H	Fl	Anti-microbial activity (Khan et al., 2015)	Clausamine H, murrayamine-A, imperatorin, phellopterin, clausenin, clausenidin, and 1-O-methylclausenolide (Houdegbe et al., 2022).	
Clusiaceae							
75. <i>Garcinia buchananii</i> Baker	Dysentery, diarrhoea, gastrointestinal issues, in cattle, sheep, goats (Graté et al., 2009)	Water extract	T	F, L Sd	Antimicrobial (Omujal et al., 2021).	Palmitic, stearic, oleic, and linoleic acids, phenolic compounds (Boakye et al., 2012).	
76. <i>Mammea africana</i> Sabine	Diarrhoea, cattle, sheep, goats, pigs, dogs, and poultry (Graté et al., 2009).	Water extract	T	B	Antibacterial, antifungal, antimalarial, anti-inflammatory, analgesic, antipyretic, and antidiarrheal (Canning et al., 2013).	Coumarins derivatives e.g., mammea A/AA, Mammea B/BB, A/AB sterol (friedelone) and xanthon (Ouahouo et al., 2004).	
77. <i>Psorospermum febrifugum</i> Spach	Mange (Graté et al., 2009).	Juice topically administered (Graté et al., 2009)	Sh	Fl	Antibacterial, antifungal, antimalarial, anti-inflammatory, analgesic, antipyretic, and antidiarrheal properties (Epifano et al., 2013).	Flavonoids, sterols (Epifano et al., 2013).	
Combretaceae							
78. <i>Combretum collinum</i> Fresen.	Diarrhoea	Water extract administered orally/topically	T	L and B	Antimicrobial, antioxidant and wound healing properties (Marquardt et al., 2020).	Triterpenes, flavonoids, lignans, and non-protein amino acids (Marquardt et al., 2020).	
79. <i>Terminalia brownii</i> Fresen.	Pink-eye in livestock	Paste	T	B	Antiviral, antioxidant, antibacterial, antifungal, anti-inflammatory, and antiparasitic (Cock, 2015).	Triterpenes, alkaloids, flavonoids, tannins, and polyphenols (Das et al., 2020).	
Commelinaceae							
80. <i>Commelina africana</i> subsp. <i>africana</i>	Body weakness in chicken (Zirintunda et al., 2024)	Pound	H	L and St	Anti-diarrheal activity (Ghosh et al., 2019)	Polyphenols, salicylic acid, p-coumaric, tannic caffeic and chlorogenic acids, 8-hydroxyquinoline, quinolones, catechol, resorcinol, flavonoids, saponins (Ghosh et al., 2019).	
81. <i>Ipomoea batatas</i> (L.) Lam.	Parturition failure	Give animal the vines to eat Crush leaves without water and smear on the vulva (Nabukenya et al., 2014).	H	L		Triterpenes/steroids, alkaloids, anthraquinones, coumarins, flavonoids, saponins, tannins, phenolic (Pochapski et al., 2011) and, oxalic acid (Faboya et al., 1983).	Oxalic acid, may crystallize as oxalate stones in the urinary tract in some people (Faboya et al., 1983)

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Table 1 (continued)

Scientific Name and (Family)	Diseases/ethnoveterinary use and animals treated	Method of preparation and use	LF	Part used	Bioactivity related to traditional uses in Ugnada	Phytochemical composition	Toxicity
82. <i>Ipomoea longituba</i> Hallier f. Crassulaceae	Low milk production	Water extract which is orally applied, yoghurt paste	H	R			
83. <i>Kalanchoe densiflora</i> Rolfe	Coccidiosis	Decoction (Bukunya-Ziraba and Kamoga, 2007)	H	Rt			
84. <i>Kalanchoe citrina</i> Schweinf.	Wounds, inflammation, in cattle, sheep, and goats, but also to dogs, chickens, and pigs (Gradé et al., 2009).	Water extract which is topically applied	H	L	Antioxidant, anti-inflammatory, and antimicrobial potential (de Andrade et al., 2023)	Glycosides, alkaloids, tannins, flavonoids, and terpenoids (de Andrade et al., 2023).	
Cucurbitaceae							
85. <i>Citrullus colocynthis</i> (L.) Schrad.	Cattle helminthosis (Matovu et al., 2020)	Grind, mix with water/macerate and drench (Matovu et al., 2020).	H	L	Antiparasitic activity (Al-Snafi, 2016)	Tannins, saponins, phenolic, flavanoids, terpenoids, alkaloids, anthranol, steroids, Cucurbitacin A, B, C, D, E (α -elaterin), caffeic acid, cardiac glycoloids (Talole et al., 2013)	Induces diarrhoea and intestinal damage in treated animals (Faraj, 1995).
86. <i>Coccinia adoensis</i> (Hochst. ex A.Rich.) Cogn.	Anaplasmosis	Squeeze liquid from flower and feed the animal	H	R	Antibacterial, anthelmintic, analgesic, antimalarial (de Boer et al., 2005).	Cucurbitacins, flavonoids, phenolic acids, saponins, tannins, and alkaloids (de Boer et al., 2005).	
87. <i>Cucumis</i> sp.	Bloat, ECF, fodder, rinder pest wounds in donkeys, cattle (Gradé et al., 2009).	Water extract with <i>Warburgia Warburgia</i> spp and sediment rock (CaCO ₃)/and <i>Momordica foetida</i> fruit juice and salt, topically (Gradé et al., 2009).	H	Fl, L	Ant-inflammatory, antifungal, antibacterial, anthelmintic, antipyretic, and wound healing (Abebe, 2022).	Alkaloids, flavonoids, glycosides, phenols, saponins, steroids, tannins, terpenoids, and cucurbitacins (Feyisa et al., 2021).	Cucurbitacins are cytotoxic (Abebe, 2022).
88. <i>Cucumis aculeatus</i> Cogn.	Ticks in livestock (Byaruhanga et al., 2015)	Fruits are crushed, mixed with water used to wash (Byaruhanga et al., 2015)	H	F		Terpenoids, anthraquinones and alkaloids (Ogutu et al., 2012).	
89. <i>Cucurbita maxima</i> Duchesne	Acute eye syndrome, dogs, horses, chickens, sheep, goats, pigs, & cattle (Gradé et al., 2009).	Paste in eye	H	St	Anti-inflammatory, antihyperlipidemic, antioxidant, and wound-healing effects (Kaźmińska et al., 2020).	Flavonoids, phenolics, carotenoids, cucurbitacins, triterpenoids, sterols, & fatty acids (Hussain et al., 2021).	
90. <i>Cucurbita moschata</i> Duchesne	Diarrhoea in chicken (Zirintunda et al., 2024)	Crushing	H	L & Sd			
91. <i>Kedrostis foetidissima</i> (Jacq.) Cogn.	Fowl typhoid	Cold water extract (Bukunya-Ziraba and Kamoga, 2007).	H	L	Antibacterial activity (Priyavardhini et al., 2008)	Phenols, alkaloids, flavonoids, tannins, terpenoids, steroids (Pavithra and Saravanan, 2020).	Hepatotoxicity by some of its compounds (Kommidi et al., 2024)
92. <i>Lagenaria sphaerica</i> (Sond.) Naudin	Constipation (Nabukunya et al., 2014), Bacillary white diarrhoea (Bukunya-Ziraba and Kamoga, 2007).	Decoction with a little rock salt and cow ghee and drench (Nabukunya et al., 2014)	H	L	Anribacterial activity (Niyonzima et al., 2024)	Tannins, phenols, saponins resins, alkaloids (Niyonzima et al., 2024)	
93. <i>Momordica foetida</i> Schumach.	Bloat, lice maggots, pinkeye, wounds, arthritits (Gradé et al., 2009), fleas, mites, mange, wounds (Nabukunya et al., 2014), fowl typhoid (Bukunya-Ziraba and Kamoga, 2007), Cattle helminthiasis (Matovu et al., 2020)	Water extract topically applied/ urine extract with juice topically smeared Juice in eye (Gradé et al., 2009; Nabukunya et al., 2014), macerate and drench 1 (Matovu et al., 2020).	T	Fl, St	Antimicrobial, anti-inflammatory, analgesic, antipyretic and antidiarrhoeal properties (Muronga et al., 2021).	Sitosteryl glucoside, 5,25-stigmastadien-3 β -yl glucoside, and 1 β -hydroxyfriedel-6-en-3-one, and cucurbitane-type triterpenoid derivatives (Muronga et al., 2021).	No pronounced toxicity against human hepatocellular & human urinary bladder carcinoma cells (Froelich et al., 2007)
Cyperaceae							
94. <i>Cyperus fischerianus</i> A. Rich.	Snake bite in donkeys, pigs, dogs, camels, and chickens	Water extract topically applied	H	R	Anthelmintic, anti-inflammatory & antibacterial (Zhang et al., 2022).	Cyperolone, cyperone, cyperotundone, quercetin, kaempferol, gallic acid, sitosterol, & linoleic acid (Babiaka et al., 2025).	
Ebenaceae							

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Table 1 (continued)

Scientific Name and (Family)	Diseases/ethnoveterinary use and animals treated	Method of preparation and use	LF	Part used	Bioactivity related to traditional uses in Ugnada	Phytochemical composition	Toxicity
95. <i>Diospyros abyssinica</i> (Hiern) F.White Euphorbiaceae	Retained placenta	Leaves and dried fruits are crushed and a decoction is drenched (Nabukenya et al., 2014)	T	L, Fr		Triterpenoids betulin, betulinic acid and lupeol (Zhong et al., 1984)	
96. <i>Alchornea cordifolia</i> (Schumach. and Thonn.) Müll.Arg.	Cattle helminthosis (Matovu et al., 2020)	Pound 2 kg of leaves plus seeds and add 1L of water, or boil (Maceration, Decoction). Drench 1/2 L to adult cattle (Matovu et al., 2020).	T	S, L	Insecticidal activity (Koomson, 2023)	Terpenes, sterols, flavonoids, tannins, glycosides, saponins alkaloids (Osadebe and Okoye, 2003)	Erratic swimming, air gulping, discolouration, loss of body equilibrium, the settlement at the bottom and death on test fishes (Essien-Ibok, 2020)
97. <i>Erythrococca bongensis</i> Pax	Intestinal parasites	Water extract is orally applied	Sh	L	Antibacterial, anti-inflammatory (Alum et al., 2025).	Flavonoids, and triterpenoids, quercetin, elabunin, mutangin, buchaninoside, canophyllol, & canophyllic acid (Alum et al., 2025).	
98. <i>Euphorbia bongensis</i> Kotschy and Peyr. ex Boiss.	Anaplasmosis, bloat contagious bovine/caprine pleuropneumonia ECF, heartwater, panacea, rinderpest in cattle, sheep, goats (Gradé et al., 2009)	Water extract which can be oral or topically applied (Gradé et al., 2009)	H	Wp, R			
99. <i>Euphorbia murielii</i> N.E.Br.	Anaplasmosis ECF, trypanosomiasis, anthrax, blackleg, and FMD (Gradé et al., 2009).	Water extract with sediment rock (CaCO ₃) Knick hide at lymph node, drip in latex (Gradé et al., 2009).	H	St ex.			
100. <i>Euphorbia cooperi</i> N.E.Br. ex A.Berger	Contagious bovine/caprine pleuropneumonia in elephants, baboons (Gradé et al., 2009)	Singe stem	H	St	Antibacterial, antifungal, antiviral, antiprotozoal, anti-inflammatory (Ahmed et al., 2019).	Dieters and triesters of 16-hydroxy-12-desoxyphorbol, a tiglane diterpene (Ahmed et al., 2019).	
101. <i>Euphorbia hirta</i> L.	Sickle cell, autoimmune conditions (Gradé et al., 2009), cough in chicken (Zirintunda et al., 2024)	Water extracts that are applied orally or topically (Gradé et al., 2009), crush (Zirintunda et al., 2024)	H	L, St	Potential for treating sickle cell disease, which could have positive effects for animals with related blood disorders. May alter immune responses, which may help treat animal autoimmune diseases (Panzu et al., 2020).	Flavonoids, tannins, terpenoids, & saponins (Ahmad et al., 2017).	
102. <i>Euphorbia heterophylla</i> L.	Helminthosis in all livestock and poultry (Gradé et al., 2009), Cattle helminthosis (Matovu et al., 2020)	Boil leaves and drench (too much causes diarrhoea) (Gradé et al., 2009), maceration and drench (Matovu et al., 2020).	H	L	Insecticidal activity (Diniz et al., 2014)	Alkaloids, Cyanide, Tannins, Flavonoids, Saponins (Smith and Krings, 2018)	Toxic to insects (Guerra, 1985)
103. <i>Euphorbia umbellata</i> (Pax) Bruyns	Theileriosis/East Coast Fever (Gradé et al., 2009 ; Nabukenya et al., 2014), ECF in cattle (Tabuti et al., 2003a)	Touch red-hot metal at lymph node, smear latex topically and inside lymph node (Gradé et al., 2009). Put sap on lymph node; irritate and dry the skin (Nabukenya et al., 2014), Sap is smeared at the swollen parts (Tabuti et al., 2003a).	Sh/ T	Ex., Ap	Anticancer, anti-inflammatory, antibacterial, and antibiofilm (Latansio de Oliveira et al., 2022).	Terpenoids, alkaloids, flavonoids, phenols, saponins, and steroids (Latansio de Oliveira et al., 2022).	
104. <i>Euphorbia tirucalli</i> L.	Enlarged lymph nodes (Gradé et al., 2009), ECF in cattle (Tabuti et al., 2003a), Newcastle disease (Zirintunda et al., 2024).	Apply latex on lymph node (Gradé et al., 2009), Apply sap (Zirintunda et al., 2024)	Sh	Ex	Analgesic, Anthelmintic, Antibacterial/antifungal/ antimicrobial, Anti-inflammatory, Antioxidant, Antiviral activity (Mali and Panchal, 2017)	Flavonoids, diterpene, steroids, alkaloids and tannins (Mali and Panchal, 2017)	

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Table 1 (continued)

Scientific Name and (Family)	Diseases/ethnoveterinary use and animals treated	Method of preparation and use	LF	Part used	Bioactivity related to traditional uses in Ugnada	Phytochemical composition	Toxicity
105. <i>Jatropha curcas</i> L.	Inflammation, wounds (Gradé et al., 2009), cattle helminthosis (Matovu et al., 2020), ECF (Tabuti et al., 2003a), NCD (Zirintunda et al., 2024), protein supplement for animal feed (Abdelgadir and Van Staden, 2013).	Apply juice topically (Gradé et al., 2009)/dry whole seeds/powder extract (Matovu et al., 2020), cut or burn swollen area and then have some sap dropped there (Tabuti et al., 2003a)	Sh	L, Fl	Antioxidant anti-inflammatory, antibacterial and antifungal (Calabrone et al., 2019), anti-inflammatory, antioxidant, antimicrobial, antiviral, analgesic, hepatoprotective, wound healing (Abdelgadir and Van Staden, 2013).	Euscaphic acid, oppositin, hoslundiol, 5-O-methylhoslundin, hosloppin, hoslundal, tectochrysin, hoslundin, hoslunfuranine and ursolic acid (Valdez-Ramirez et al., 2023), terpenes, phorbol esters, cyclic peptides, alkaloids, lignans, coumarins and phytosterols (Abdelgadir and Van Staden, 2013).	The aqueous and ethanolic leaf extracts had LD ₅₀ values ≤ 2000 mg/kg bw in rats (Valdez-Ramirez et al., 2023), antifertility activity, vomiting and diarrhoea (Abdelgadir and Van Staden, 2013)
106. <i>Acalypha fruticosa</i> Forssk	Pneumonia in cattle, goats (Gradé et al., 2009),	Water extract given orally in goats and cattle (Gradé et al., 2009),	Sh	R, L	Antimutagenic, antiinflammatory, antimalarial, anthelmintic, and antimicrobial (Zafar et al., 2025).	Phenolics e.g., caffeic, ferulic acid, and gallic acid, and quercetin, flavonoids, saponins, alkaloids, tannins, terpenoids, and steroids (Fawzy et al., 2017).	Non toxic to 4 human cancer cell lines (SK-MEL, KB, BT- 549, SK-OV-3) (Fawzy et al., 2017).
107. <i>Tragia brevipes</i> Pax	Diarrhoea, mastitis, skin infections, and eye diseases (Rahman et al., 2023), Cattle helminthosis (Matovu et al., 2020).	Powdered and given orally or topically (Rahman et al., 2023),/ maceration and given to cattle	H	L, R	Anti-microbial activity (Migabo et al., 2023).	Flavonoids, saponins, glycosides, phenols, tannin, and resins (Migabo et al., 2023).	
108. <i>Ricinus communis</i> L.	Skin infections, gastrointestinal issues, and inflammation Helminthosis in all livestock and poultry (Matovu et al., 2020), NCD (Zirintunda et al., 2024).	Water extracts administered orally and topically. Pound seeds (powdered extract) and put into the mouth (Matovu et al., 2020).	H	L, St and Sd	Antimicrobial, (Abbas et al., 2018), insecticidal, antinociceptive, anti-inflammatory, analgesic, and anticonvulsant activity (Abdul et al., 2018)	Alkaloids, flavonoids, terpenes, saponins, phenolic compounds such as kaempferol, gallic acid, ricin, rutin, lupeol, ricinoleic acid, pinene, thujoneand genticic acid (Abdul et al., 2018)	Ricin causes cell death and ricinine causes poisoning (Bozza et al., 2015)
Fabaceae							
109. <i>Abrus precatorius</i> L.	Earache in cattle, rabies diarrhoea in goats and, NCD (Gradé et al., 2009). Cattle helminthosis (Matovu et al., 2020), cataract in cattle (Tabuti et al., 2003a).	Powder in ears, decoction, (Gradé et al., 2009), maceration and or drench (Matovu et al., 2020),	H	St, Sd, R, L	Antimicrobial, antiinflammatory and anti-oxidant (Okhale and EM, 2016).	Alkaloids, terpenoids, and flavonoids, e.g., luteolin, abrectorin, orientin, isoorientin, glycyrrhizin, abrusoside, abrusogenin, and abruquinones (Okhale and EM, 2016).	Toxic protein abrusin which inhibit protein synthesis (Kaula et al., 2022).
110. <i>Vachellia tortuosa</i> (L.) Seigler & Ebinger	Dewormer, rinderpest, lumpy skin disease in cattle (Gradé et al., 2009), diarrhoea in chicken (Zirintunda et al., 2024)	Directly fed to animals as fodder (Gradé et al., 2009)	Sh	Fl	Antimicrobial, anti-parasitic, antiinflammatory antidiabetic (Batiha et al., 2022).	Phenols, steroids, terpenoids, flavonoids, saponins, glycosides, quercetin, rutin, ferulic, betulinic and caffeic acid (Batiha et al., 2022; León De Pinto et al., 1998).	
111. <i>Vachellia abyssinica</i> (Hochst. ex Benth.) Kyal. & Boatwr.	Lumpy skin disease and snake bite in sheep, cattle and goats (Gradé et al., 2009)	Water extract administered topically in sheep, goats and cattle (Gradé et al., 2009)	T	B	Antiinflammatory, anti-microbial, & antivenom and antiviral (Adhikari and Rangra, 2023).	Tannins, steroids, saponins (Adhikari and Rangra, 2023).	
112. <i>Senegalia brevispica</i> (Harm) Seigler & Ebinger	Deworming in sheep, goats and cattle (Gradé et al., 2009)	Direct as fodder (Gradé et al., 2009).	Sh	Wp	Antiinflammatory, antipyretic, antidiarrhoeal, antimicrobial, antifungal, antihelminthic (Batiha et al., 2022).	Flavonoids, tannins, saponins, alkaloids, and terpenoids (Batiha et al., 2022).	
113. <i>Vachellia drepanolobium</i> (Harms ex Y. Sjöstedt) P.J.H. Hurter	Diarrhoea in cattle, goats and dogs (Gradé et al., 2009), ticks in livestock, (Byaruhanga et al., 2015)	Water extract administered topically in goats and cattle (Gradé et al., 2009), Fresh roots are crushed and used for washing and drench (Byaruhanga et al., 2015).	Sh	R	Antimicrobial, antifungal, antiprotozoal, anti-inflammatory, antidiarrhoeal (Mollel et al., 2020).	Quercetin, kaempferol, rutin, catechin, epicatechin, gallic, caffeic acid, chlorogenic acid, ferulic acid, and β-sitosterol (Batiha et al., 2022).	Low acute toxicity in mice: oral LD ₅₀ = 3.9 g/kg (Batiha et al., 2022).
114. <i>Vachellia gerrardi</i> (Benth.) P.J.H. Hurter	Cough in cattle (Gradé et al., 2009), ticks in livestock (Byaruhanga et al., 2015)	Water extract is orally administered in cattle (Gradé et al., 2009), Fresh leaves are crushed, mixed with water, and the extract	Sh	B	Antiinflammatory, antioxidant, antidiarrhoeal, antiviral, and hepato protective in mice (Batiha et al., 2022).	Quercetin, kaempferol, rutin, catechin, epicatechin, gallic, caffeic, chlorogenic &, ferulic acid and β-sitosterol (Elkahoui et al., 2024; Raweh et al., 2022).	Low acute toxicity in mice, with an oral LD ₅₀ of 3.9 g/kg (Batiha et al., 2022).

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Table 1 (continued)

Scientific Name and (Family)	Diseases/ethnoveterinary use and animals treated	Method of preparation and use	LF	Part used	Bioactivity related to traditional uses in Ugnada	Phytochemical composition	Toxicity
		is used. Washing animal body with extract					
115. <i>Vachellia amythetophylla</i> (Steud. ex A. Rich.) Kyal. & Boatwr.	Lumpy skin disease, skin disease with intestinal adhesions in cattle and sheep (Gradé et al., 2009)	Water extract administered topically and orally in cattle and sheep (Gradé et al., 2009)	Sh	B, L			
116. <i>Dalbergia melanoxyton</i> Guill. & Perr	Helminthosis, ticks in livestock (Byaruhanga et al., 2015).	Crush fresh roots and bark, mixed with water, and drench (Byaruhanga et al., 2015).	Sh	R, B		Neoflavonoids, benzofuran (Lin et al., 2020), tannins, saponins, sterols (Chalo et al., 2023; Najeeb et al., 2018).	
117. <i>Senegalia mellifera</i> (Vahl) Seigler & Ebinger	Otitis secondary to biting ticks (Gradé et al., 2009)	Water extract applied topically (Gradé et al., 2009)	Sh	B	Antimicrobial, insecticidal (Mutai et al., 2009).	Tannins, phenolics, saponins, terpenoids, alkaloids, flavonoids, (Mutai et al., 2009).	
118. <i>Vachellia oerfota</i> (Forssk.) Kyal. and Boatwr.	Pneumonia, rinderpest, trypanosomiasis (Gradé et al., 2009).	Water extract given orally, directly fed to animals as fodder (Gradé et al., 2009)	Sh	B, R	Antimicrobial, antifungal, antiprotozoal, anti-inflammatory, antidiabetic (Tiwari et al., 2023).	Phenolics especially quercetin, kaempferol, rutin, and gallic acid (Akapo et al., 2025).	
119. <i>Vachellia hockii</i> (De Wild.) Seigler & Ebinger	NCD (Zirintunda et al., 2024)	Pound	Sh	L, B			
120. <i>Senegalia senegal</i> (L.) Britton	Delivery and post partum pain in sheep, goats and cattle (Gradé et al., 2009)	Directly as fodder, water extract orally ingested in sheep, goats and camels (Gradé et al., 2009).	Sh	B, Fl, L	Antimicrobial, antifungal, anti-inflammatory, anthelmintic (Djambo et al., 2025; Msimango et al., 2025).	Polysaccharides and glycoproteins (Uzunugbe et al., 2019).	Low toxicity in mice No deaths/behavioural changes observed at 5000 mg/kg bw; LD ₅₀ = 5000 mg/kg bw (Magnini et al., 2021).
121. <i>Vachellia seyal</i> (Delile) P.J.H. Hurter	Diarrhoea, lumpy skin disease in livestock e.g., sheep, goats and cattle (Gradé et al., 2009), ECF in cattle (Tabuti et al., 2003a)	Water extract both topically and orally administered in goats, sheep and cattle (Gradé et al., 2009)	Sh	R B, L	Antimicrobial, antifungal, anti-inflammatory, antidiabetic, and antioxidant activities in sheep, goats & cattle (Elmi et al., 2020), analgesis, atinflammatory (Kedir et al., 2024).	Polysaccharides and glycoproteins (Ashour et al., 2022).	Safe at 2000 mg/kg in both rats & mice (Kedir et al., 2024).
122. <i>Vachellia tortilis</i> subsp. <i>spirocarpa</i> (Hochst. ex A. Rich.) Kyal. & Boatwr.	Abscess in goats and goat pox	Water extract topically administered in goats and various livestock (Prayogo et al., 2021)	Sh	L	Antimicrobial, anti-inflammatory, & antioxidant activities (Prayogo et al., 2021).	Catechin, quercetin, and kaempferol (More et al., 2021; Taha et al., 2022).	Minimal acute toxicity in mice, LD ₅₀ = 5.2 g/kg bw (Batiha et al., 2022).
123. <i>Vachellia sp</i>	Goat pox, retained placenta, low milk production (Gradé et al., 2009)	Water extract with <i>Ocimum basilicum</i> (Gradé et al., 2009).	Sh	Fl, Lv	Antimicrobial, anti-inflammatory, (Jaeger et al., 2019).	Tannins, flavonoids, alkaloids, saponins, terpenoids, steroids, and phenols (More et al., 2021)	Low acute toxicity in mice, LD ₅₀ of 4.8 g/kg bw (Jaeger et al., 2019).
124. <i>Albizia amara</i> (Roxb.) Boivin	Abscess, chronic wound, Mange hoof disease, maggots, otitis with pus, often ticks (Gradé et al., 2009)	Poultry, oil extraction with topical administration (Gradé et al., 2009)	Sh	L	Antiinflammatory, analgesic, antipyretic, antifungal, antibacterial, and wound healing (Njoroge et al., 2010).	Flavonoids, saponins, tannins, alkaloids, steroids, terpenoids, phenols, glycosides. quercetin, kaempferol, rutin, lupeol, betulinic acid, and albigenin (Srivastava et al., 2020).	Minimal acute and subacute toxicity in mice and rats (Srivastava et al., 2020).
125. <i>Albizia anthelmintica</i> (A. Rich.) Brongn.	Bloat and cough, intestinal parasites in goats, sheep and cattle (Gradé et al., 2009)	Water extract administered both topically as poultry and orally in sheep, goats and cattle (Gradé et al., 2009)	Sh	R, B	Antiparasitic, antinematodal, antibacterial against (El Khodary et al., 2021).	Alkaloids, flavonoids, glycosides, phenolics, saponins, steroids, tannins, and terpenoids (Mohamed et al., 2013).	A dose rate of 1000 mg/kg b.w resulted death of all mice (Githiori et al., 2003).
126. <i>Albizia coriaria</i> Welw. ex Oliv.	Barreness, pneumonia, increase fertility, fever, contagious bovine/caprine pleuropneumonia (Gradé et al., 2009), bacillary white	Water extract given orally in cows (Gradé et al., 2009), decoction (Bukanya-Ziraba and Kamoga, 2007), fresh leaves and bark	T	B, L, R	Antioxidant, antibacterial, antifungal, antiinflammatory, antiparasitic, and anti-giardial (Omara et al., 2023).	Lupeol, luponone, betulinic acid, betulin, and benzyl alcohol (Omara et al., 2023).	Minimal acute toxicity in mice, LD ₅₀ = 3.2 g/kg (Omara et al., 2023). Highly cytotoxic to U87 cells (Anywar, 2022; Anywar et al., 2021).

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Table 1 (continued)

Scientific Name and (Family)	Diseases/ethnoveterinary use and animals treated	Method of preparation and use	LF	Part used	Bioactivity related to traditional uses in Ugnada	Phytochemical composition	Toxicity
127. <i>Albizia zygia</i> (DC.) J.F.Macbr.	diarrhoea, fowl typhoid (Bukunya-Ziraba and Kamoga, 2007), helminthosis in livestock (Byaruhanga et al., 2015), ECF (Tabuti et al., 2003a), NCD & diarrhoea (Zirintunda et al., 2024) Painful red eyes, lice, Mange in livestock (Gradé et al., 2009)	crushed and used for drenching (Byaruhanga et al., 2015), decoction/infusion with <i>Oncoba spinosa</i> /roots of <i>Milicia excelsa</i> and <i>S.longipedunculata</i> (Tabuti et al., 2003a). Poultice topically administered in goats, sheep and cattle (Gradé et al., 2009).	T	L	Antiinflammatory, analgesic, in chicks, rats and mice (Abotsi et al., 2017).	Lupen-20(30)-3b-ol, pyrogallol, phenol, phytol, dl- α -tocopherol, oleamide, campesterol, stigmasterol, β -sitosterol and lupeol (Olukanni, 2020).	No acute toxicity in mice at 5000 mg/kg (Abotsi et al., 2017).
128. <i>Chamaecrista nigricans</i> (Vahl) Greene	Abscess, dermatophilosis, fever, intestinal parasites, meningitis, wounds mainly in goats, cattle, sheep (Gradé et al., 2009)	Powder topically applied/water extract with <i>P. thonningai</i> bark, used to wash (Gradé et al., 2009)	Sh	L	Antimicrobial, antiulcer, analgesic, anti-inflammatory, insecticidal, and larvicidal activities (Ayo, 2010).	Emodic acid, citreoresein, and other anthraquinone derivatives, luteolin, flavonoid, and hydroxyestraneic acid (Ayo et al., 2009).	bis (2- ethylhexyl) phthalate, one of its secondary compounds, is toxic to humans (Ayo et al., 2009).
129. <i>Crotalaria</i> sp.	Poisonous gives cattle foot rot anaplasmosis (Gradé et al., 2009)	Cattle walk on mature pods in pasture (Gradé et al., 2009)	H	Fl	Antioxidant, antibacterial, antifungal, anti-inflammatory, antinociceptive, antimalarial & antitumor (Saba, 2019).	Flavonoids, phenols, tannins, saponins, terpenoids, and steroids, pyrrolizidine alkaloids e.g., senecionine, retrorsine, senkirkine, and monocrotaline (Saba, 2019).	
130. <i>Erythrina abyssinica</i> Lam.	Deworming in cattle, goats, and sheep and reduces inflammation (Gradé et al., 2009), Fowl typhoid (Bukunya-Ziraba and Kamoga, 2007), Conjunctivitis (Matovu et al., 2020), cough and ectoparasites & infectious bronchitis in chicken (Zirintunda et al., 2024)	Decoctions or infusions (Bukunya-Ziraba and Kamoga, 2007; Gradé et al., 2009), pound about a handful (200g) of the flowers (Maceration). Squeeze 4–5 drops into the eye (Matovu et al., 2020).	T	B, L, R, F	Antibacterial, antifunga l (Obakiro et al., 2021).	Phenols, flavonoids, saponins, alkaloids, tannins (Obakiro et al., 2021).	Minimal toxicity on the liver, kidney, and hematopoietic tissue (Obakiro et al., 2021).
131. <i>Phaseolus vulgaris</i> L.	Fowl pox in chicken (Zirintunda et al., 2024)	Pound	H	L, S, R			
132. <i>Piliostigma thonningii</i> (Schumach.) Milne-Redh.	Dermatophilosis, bloody diarrhoea, trypanosomiasis in cattle, sheep, goats (Gradé et al., 2009),	Poultice (Gradé et al., 2009),	T	B	Antimicrobial, antiviral, antimalarial, anti-inflammatory (Abubakar et al., 2024; Nethathe et al., 2025).	Terpenoids, alkaloids, phenolics, flavonoids, saponins, tannins, and steroids, quercetin, kaempferol, piliostigmin (Abubakar et al., 2024; Nethathe et al., 2025).	
133. <i>Philenoptera laxiflora</i> (Guill. & Perr.) Roberty	Cough and diarrhoea in chicken (Zirintunda et al., 2024)	Pound bark (Zirintunda et al., 2024)	T	Bark			
134. <i>Pseudarthria confertiflora</i> (A. Rich.) Baker	Helminthosis in all livestock and poultry (Nabukunya et al., 2014).	Boil leaves or bark for 1 h, cool and drench (Nabukunya et al., 2014).	H	L, B			
135. <i>Pseudarthria hookeri</i> Wight & Arn.	Cattle helminthosis (Matovu et al., 2020)	Maceration and (Matovu et al., 2020).	Sh			Alkaloids, flavonoids, terpenoids, peptide, polysaccharide and saponins (Yaya et al., 2023).	
136. <i>Senna occidentalis</i> (L.) Link	Worms in cattle, goats, sheep andpoultry, Helminthosis in all livestock and poultry, constipation, heart problems, rotten wounds in mouth in cattle, sheep, goats and horses (Gradé et al., 2009), singamus (Bukunya-Ziraba and Kamoga, 2007; Matovu et al., 2020), fiarrhoea and cough in cattle (Tabuti et al., 2003a)	Decoct and drench (Gradé et al., 2009; Matovu et al., 2020), decoction (Bukunya-Ziraba and Kamoga, 2007),	Sh	St, B, R, Fl, L	Antibacterial, antifungal, anthelmintic, anti-inflammatory (Suleiman et al., 2014).	Neolignan, flavonoidal glycosides, flavonolignans, glucopyranoside, betulinic and vanillic acid (Suleiman et al., 2014), sennosides (Mogaka et al., 2023), alkaloids, tannins, saponins, phytosterols, phenols, flavonoids (Suleiman et al., 2014)	Poisoning and death acute atrophy of the liver and muscles (Delachiaive and De Pinho, 2003)

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Scientific Name and (Family)	Diseases/ethnoveterinary use and animals treated	Method of preparation and use	LF	Part used	Bioactivity related to traditional uses in Ugnada	Phytochemical composition	Toxicity
137. <i>Senna</i> Spp	Fever, rotten wounds in mouth, pneumonia (Nabukenya et al., 2014).	Water extract, Water infusion with salt, wash mouth (Nabukenya et al., 2014).	H	L			
138. <i>Senna didymobotrya</i> (Fresen.) H.S. Irwin & Barneby	Anaplasmosis, ECF, and ectoparasites, diarrhoea (Nabukenya et al., 2014), helminthosis (Matovu et al., 2020).	Water extract (Matovu et al., 2020), maceration, decoction and drench (Matovu et al., 2020).	Sh	R	Antimicrobial, anthelmintic (Nyamwamu et al., 2015), analgesic, insecticidal, antiinflammatory (Orwa and Njue, 2019).	Anthroquinones, alkaloids, terpenoids, anthraquinones, tannins, saponins, phenols and flavonoids (Orwa and Njue, 2019)	Slightly toxic. Harmful to both aquatic invertebrates and fish (Nyamwamu et al., 2015).
139. <i>Senna hirsuta</i> (L.) H.S.Irwin & Barneby	Helminthosis in all livestock and poultry (Nabukenya et al., 2014).	Boil dry leaves with little rock salt and drench or crush fresh leaves and drench (Nabukenya et al., 2014).	Sh	S, L			
140. <i>Senna siamea</i> (Lam.) H.S.Irwin & Barneby	Cough in chicken (Zirintunda et al., 2024)	Crushing (Zirintunda et al., 2024).	T	L	Antibacterial activity (Ogunniran et al., 2024)	Saponin, alkaloids, antraquinones and phylobatannins (Smith, 2009)	Alkaloids are often toxic to man and may have dramatic physiological activities (Shelton, 1980)
141. <i>Tamarindus indica</i> L.	Inapetance in calves with diarrhoea, FMD, rotten wounds in mouth (Gradé et al., 2009), coccidiosis, fowl typhoid (Bukunya-Ziraba and Kamoga, 2007), NCD and bacterial infections in chicken (Zirintunda et al., 2024)	Infusion (Gradé et al., 2009), Decoction with unripe fruit (Bukunya-Ziraba and Kamoga, 2007), pound (Zirintunda et al., 2024)	T	Fl Fr	Anthelmintic, antibacterial, wound healing, antiinflammatory, antiviral, analgesis, antifungal (Tugume et al., 2020).	Anthocyanins, flavonoids, phenolics, antibacterial (Tugume et al., 2020).	Cytotoxic to liver tumor cells (Iskandar et al., 2017).
142. <i>Tephrosia vogelii</i> Hook.f.	Ticks (Gradé et al., 2009), insecticide/acaricide (Siame et al., 2019), chicken lice, mites (Bukunya-Ziraba and Kamoga, 2007), skin disease 'lukuku' and wounds in cattle and maggot infested wounds (Tabuti et al., 2003a).	Water extracts, applied topically (Gradé et al., 2009), decoction (Bukunya-Ziraba and Kamoga, 2007), rubbed on animal skin, sap applied to wounds (Tabuti et al., 2003a).	H	L	Insecticidal activity (Shin-Foon, 1989)	Deguelin, tephrosin, and rotenone (Belmain et al., 2012).	Rotenone is toxic causing foetotoxicity and death of offspring in genuine pigs at doses of 4.5 and 9.0 mg/kg/day (Moriya et al., 1983).
143. <i>Tylosema fassoglense</i> (Kotschy ex Schweinf.) Torre & Hillc.	Fowl typhoid (Bukunya-Ziraba and Kamoga, 2007), ticks in livestock (Byaruhanga et al., 2015)	Dedcoction (Bukunya-Ziraba and Kamoga, 2007), fresh leaves and stem are crushed with water and used for washing animal (Byaruhanga et al., 2015).	H	Rt, L. St	Antibacterial activity (Mukavi et al., 2020)	Polyphenols, flavonoids, tannins, and alkaloids (Kudamba et al., 2022)	
144. <i>Tephrosia</i> Spp.	Abscess, anaplasmosis, fever (Gradé et al., 2009),	Water extracts, applied topically	H	L			
145. <i>Neorautanenia mitis</i> (A. Rich.) Verdc.	Fleas, lice, ficks (Gradé et al., 2009),	Water extract, topically	H	R		Alkaloids, phenols, terpenoids, flavonoids, and tannins, rautandiol A & B (Kagisha et al., 2020).	
146. <i>Indigofera spicata</i> Forssk.	Appetite stimulant, calf diarrhoea, trypanosomiasis (Gradé et al., 2009),	Water extract is orally applied (Gradé et al., 2009).	H	L	Analgesis & anti-inflammatory (Birhane and Asres, 2014).	Benzofurans, flavonoids, saponins triterpenes, steroids, phthalates, fatty acids (Mouafon et al., 2021), indospicine (Ossedryver et al., 2013).	Leaf & seed hepatotoxic to rabbits & mice (Hutton et al., 1958). Causes ataxia, paresis, depression, muscle fasciculations, dysphagia, ptyalism & halitosis (Ossedryver et al., 2013). Idospicine residues accumulate & persist in bovine tissues (Fletcher et al., 2016).
Lamiaceae							
147. <i>Aeollanthus</i> sp	Trypanosomiasis, diarrhoea in sheep and goats, wound healing, skin infections (Gradé et al., 2009).	Water extract administered orally in sheep and goats (Gradé et al., 2009).	H	R	Antimicrobial, insecticidal (Martins et al., 2016).	Flavonoids, phenolic, terpenoids, essential oils (Martins et al., 2016).	The oil is generally recognized as safe at the experimental dosage. Did not cause any mortality in the experimental rats at 2000 mg/kg bw (Yovo et al., 2019).
148. <i>Clerodendrum rotundifolium</i> Oliv.	Cattle helminthosis (Matovu et al., 2020)	Maceration/decoction and drench (Matovu et al., 2020).	H	L		Tannins, saponins, flavonoids and cardiac glycosides (Adia et al., 2016)	

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Table 1 (continued)

Scientific Name and (Family)	Diseases/ethnoveterinary use and animals treated	Method of preparation and use	LF	Part used	Bioactivity related to traditional uses in Ugnada	Phytochemical composition	Toxicity
149. <i>Rotheca myricoides</i> (Hochst.) Steane & Mabb.	ECF in cattle (Tabuti et al., 2003a)	Infusion (Tabuti et al., 2003a)	Sh/T	L	Antifeedant activity (Njeru et al., 2016)	Alkaloids, phenols, anthraquinones, terpenoids, and flavonoids (Njeru et al., 2016)	
150. <i>Fuerstia africana</i> T.C.E.Fr.	Cattle helminthosis (Matovu et al., 2020),	Decocotion/maceration of roots and leaves and drench (Matovu et al., 2020)	Sh	L, S, R		Terpenoids (Eugster, 1980).	
151. <i>Rotheca myricoides</i> (Hochst.) Steane & Mabb.	Bovine Ephemeral Fever, retained placenta, diarrhoea, dysentery, malaria, fever, cough, respiratory problems, intestinal worms, and wounds (Gradé et al., 2009),	Pounded roots decoct and drench (Gradé et al., 2009),	Sh	R	Antiviral activity (Litabe et al., 2024)	Alkaloids, phenols, anthraquinones, terpenoids, and flavonoids (Njeru et al., 2016)	
152. <i>Hoslundia opposita</i> Vahl	Mastitis, infertility, abortion, retained placenta, diarrhoea, dysentery, malaria, fever, cough, respiratory problems, intestinal worms, and wounds (Gradé et al., 2009),.	Powder in eye (Gradé et al., 2009)	H	L	Immunomodulatory, anti-inflammatory, antifungal, antiviral, antihelminthic and antiprotozoal (Namuga et al., 2024; Oyeyemi et al., 2023)	Diterpenoids 3-O-benzoylhosloppone and 3-O-cinnamoylhosloppone, triterpenoid euscaphic and ursolic acid (Namuga et al., 2024; Oyeyemi et al., 2023).	Elevated AST and ALT in high doses when administered to mice (Oloyede, 2020)
153. <i>Leonotis nepetifolia</i> (L.) R. Br.	Intestinal parasites, maggots (Gradé et al., 2009),	Water extract/poultice (Gradé et al., 2009).	H	Wp, L	Anti-microbial, anti-inflammatory (Gang and Kang, 2022).	Alkaloids, phenolics, flavonoids, tannins, steroids, glycosides, coumarins, anthocyanins, saponins, hydroxynepetaefolin, nepetaefuran, leonotinin, leonepetaeofolin A-E (Gang and Kang, 2022).	Pronounced embryotoxic and teratogenic effects in zebrafish (Adolpho et al., 2025).
154. <i>Ocimum gratissimum</i> subsp. <i>Gratissimum</i>	Calf diarrhoea, mosquito control (Gradé et al., 2009)	Water extract Smudge in fire, and/or hang near kraal (Gradé et al., 2009)	H	Rt Wp	Antimicrobial, analgesic, antipyretic, anti-inflammatory, antidiarrheal, (Nguta and Kiraithe, 2019)	Tannins, flavonoids, saponins, steroids, alkaloids, and phenol	Moderate cytotoxicity against various human cancer cell lines (Caamal-Herrera et al., 2018).
155. <i>Tetradenia riparia</i> (Hochst.) Codd	Gastrointestinal issues, skin diseases, and respiratory problems in cattle, sheep, goats, helminthosis in all livestock and poultry (Nalule et al., 2011), ECF (Tabuti et al., 2003a).	Infusions,/crush leaves, and put on the wound (Nalule et al., 2011), infusion to which salt is added and given to calf (Tabuti et al., 2003a).	Sh	Fl, St	Antimicrobial, antioxidant, analgesic, anti-leishmania, anti-tuberculosis and anti-parasitic (Panda et al., 2022)	Terpenes, fenchone, 14-hydroxy-9-epi (E)-caryophyllene, 9β, 13β-epoxy-7-abetene, and 6,7-dehydroroileanone (Panda et al., 2022).	Severe toxic inflammatory response of mucous membranes at all body orifices, profuse salivation in man (Panda et al., 2022).
156. <i>Thymus vulgaris</i> L.	Body weakness in chicken (Zirintunda et al., 2024)	Crush	H	L	Antibacterial activity (Mukhtar et al., 2018)	Phenolic compounds, terpenoids, flavonoids, steroids, alkaloids, tannins, and saponin (Patil et al., 2021)	2000 mg/kg bw caused the death of 2/3 within 24 h preceded by rapid onset of toxicity signs in the respiratory, CNS and autonomic systems (Rojas-Armas et al., 2019)
Meliaceae							
157. <i>Azadirachta indica</i> A.Juss	Chicken mites, eye problems, fleas, genital diseases, Mange, mosquito control, wounds, gastrointestinal disorders, pneumonia, parasitic worms, snake bites (Gradé et al., 2009), BWD, NCD, fowl typhoid, bacillary white diarrhoea (Bukeny-Ziraba and Kamoga, 2007), helminthosis (Byaruhanga et al., 2015), skin disease (itching) in cattle (Tabuti et al., 2003a), NCD, diarrhoea and shistomoniasis in chicken (Zirintunda et al., 2024)	Water extract, grind seeds to extract oil, apply topically in poultry and cattle/decoction administered orally (Gradé et al., 2009)/fresh crushed, mixed with water and used for drenching. Fruits are dried, crushed into powder, and then boiled to extract an oil suspension which is smeared on the animal (Byaruhanga et al., 2015).	T	Lv, Sd, Rt	Antiparasitic, antifungal, antibacterial, antiviral, anti-inflammatory, antipyretic, antidiarrhoeal, and insecticidal (Njoga et al., 2022; Sarkar et al., 2021).	Terpenoids, flavonoids, phenolics, alkaloids, and glycosides (Uchegbu et al., 2011); nimbolide, azadirachtin, and gedunin (Sarkar et al., 2021).	Toxicity depends on used extraction method, dosage, administration route, animal species, and physiological condition (Tamilvanan Sujatha et al., 2017). Generally considered safe (Njoga et al., 2022).
158. <i>Turraea floribunda</i> Hochst.	Panacea (Gradé et al., 2009)	Water extract (Gradé et al., 2009)	T	B			
Loganiaceae							

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Table 1 (continued)

Scientific Name and (Family)	Diseases/ethnoveterinary use and animals treated	Method of preparation and use	LF	Part used	Bioactivity related to traditional uses in Ugnada	Phytochemical composition	Toxicity
159. <i>Strychnos innocua</i> Delile	ECF in cattle (Tabuti et al., 2003a)	Infusion 300–500 ml.	Sh	R		α -Cubebene, Dibutyl benzene –1,2 – dicarboxylate, β -Elemenone, 4-Methoxy-2- nitroformanilide, Azelaic acid, Glycidol stearate, Chloromethyl 2-chlorodecanoate, γ -terpinene (Alagbe, 2023)	
Malvaceae							
160. <i>Abelmoschus ficulneus</i> (L.) Wight and Arn.	Newcastle disease (Zirintunda et al., 2024)	Crush (Zirintunda et al., 2024)	H	L		Flavonoids (Dashputre and Bandawane, 2021)	
161. <i>Abutilon hirtum</i> (Lam.) Sweet	Retained placenta and anaplasmosis in cattle, dewormer cattle, sheep, and goats (Gradé et al., 2009)	Water extract administered orally in goats, sheep and cattle (Gradé et al., 2009)	Sh	B		Mucilage, tannins, β -sitosterol, asparagines, flavonoids, alkaloids and sesquiterpenes (Gomaa et al., 2022).	Cytotoxic at 1.07 mg/g (Hamed et al., 2017)
162. <i>Grewia bicolor</i> Juss.	Retained placenta (Gradé et al., 2009)	Water extract which is orally applied (Gradé et al., 2009)	T	B	Antibacterial, antidiabetic, anti-inflammatory, and anti-malarial (Masisi et al., 2021).	Flavonoids, phenolics, tannins, terpenoids, steroids, alkaloids, and saponins, gallic acid, ursolic acid, myricetin, kaempferol, and quercetin (Masisi et al., 2021).	
163. <i>Grewia mollis</i> Juss.	Retained placenta Constipation (Gradé et al., 2009).	Water extract used orally or by enema (Gradé et al., 2009)	T	B	Anti-inflammatory, antioxidant, antidiabetic, antibacterial, and wound-healing qualities	Phenols, tannins, saponins, alkaloids, and steroids, glucuronic acid.	
164. <i>Grewia villosa</i> Willd.	Retained placenta (Gradé et al., 2009)	Water extract which is topically (Gradé et al., 2009)	T	B	Antibacterial (Mummed et al., 2018).	Alkaloids, saponins, tannins, phenols, icosane, docosane, tricosane, bis(2-ethylhexyl) decanedioate, icosane (Kamau et al., 2025).	
165. <i>Hibiscus trionum</i> L.	Fodder, diarrhoea in goats and cattle (Gradé et al., 2009)	Direct as fodder (Gradé et al., 2009)	H	St	Improves weight loss caused by experimental diabetes in rats, antidiabetic, hypolipidemic, hepatoprotective (Ziyanok-Demirtas, 2024).	Cleomiscosin A, mansonone (Kamel et al., 2024).	Toxic in goats, clinical signs included inappetence, salivation, abdominal pain, diarrhoea, weakness of hind limbs cardiac & renal haemorrhages etc (Onsa et al., 2007)
166. <i>Sida cuneifolia</i> Roxb.	Fractures and sprains (Gradé et al., 2009)	Decoction and infusions applied topically or orally (Gradé et al., 2009)	Sh	L, R, Wp		Alkaloids, flavonoids, and coumarin (Aminah et al., 2021)	
167. <i>Theobroma cacao</i> L.	Diarrhoea in chicken (Zirintunda et al., 2024)	Crush seeds	Sh	Sd	Antibacterial property (Patil et al., 2022)	Phenols, flavonoids (catechin, epicatechin, and procyanidins) (Akinmoladun et al., 2016)	
168. <i>Sida ovata</i> Forssk.	Fodder, retained placenta (Gradé et al., 2009)	Water extract (Gradé et al., 2009)	H	L, Fl	Antimicrobial, inflammation and analgesic and anti-inflammatory.	Terpenoids, alkaloids, flavonoids, phenols, saponins and steroids.	
Meliaceae							
169. <i>Melia azedarach</i> L.	Contagious bovine/caprine pleuropneumonia, measles, Rinderpest in cattle, sheep, goats, pigs, dogs, and poultry (Gradé et al., 2009), body weakness, wounds, infectious Coryza in chicken, chronic respiratory disease, fowl pox and fowl cholera, ectoparasites, infectious bronchitis in chicken (Zirintunda et al., 2024).	Water extract (Gradé et al., 2009),	T	R, B	Antioxidant, antimicrobial, antiviral, antifeedant, antifungal, and cytotoxic effects (Hieu et al., 2022)	Coumarin derivatives, flavonoids, sterols, and xanthenes (Doughnon and Ito, 2022)	Poisoning may result in gastrointestinal, respiratory, cardiovascular, neurological effects, and death in severe cases (Phua et al., 2008).
170. <i>Trichilia prieureana</i> A. Juss.	Chest pain, East coast fever (Gradé et al., 2009), coccidiosis (Zirintunda et al., 2024).	Water extract (Gradé et al., 2009), pound and give (Zirintunda et al., 2024).	T	B	Antimicrobial (da Silva et al., 2021).	Terpenes, limonoids (Resetar et al., 2023).	Mild hepatotoxicity on prolonged administration (Roland et al., 2022).
Menispermaceae							

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Table 1 (continued)

Scientific Name and (Family)	Diseases/ethnoveterinary use and animals treated	Method of preparation and use	LF	Part used	Bioactivity related to traditional uses in Ugnada	Phytochemical composition	Toxicity
171. <i>Chasmanthera dependens</i> Hochst.	Anaplasmosis, fever East Coast fever, diarrhoea, Foot and mouth disease prophylaxis Cattle and goats (Gradé et al., 2009).	Water extract which is orally applied/with <i>T. indica</i> fruits for FMD (Gradé et al., 2009)	H	St, R, B	Antiplasmodial, antioxidant, analgesic, anti-inflammatory, antihyperlipidemic & antimicrobial (Kola-Mustapha et al., 2020).	Alkaloids, flavonoids, saponins, tannins, terpenoids, steroids, glycosides, 4-hydroxyanthraquinone-2-carboxylic acid & β -sitosterol acetate (Enenebeaku et al., 2021).	Minimal cytotoxicity against brine shrimp larvae & no chronic toxicity in experimental rats (Kola-Mustapha et al., 2020).
172. <i>Hyalosepalum afrum</i> (Miers) Troupin	Anaplasmosis (Gradé et al., 2009).	Water extract (Gradé et al., 2009)	T	R			
Moraceae							
173. <i>Artocarpus heterophyllus</i> Lam.	Newcastle disease in chicken (Zirintunda et al., 2024)	Crush (Zirintunda et al., 2024)	T	B	Antibacterial effects (Baliga et al., 2011)	Carotenoids, flavanoids, volatile acids, sterols and tannins (Arung et al., 2007)	Low acute toxicity with zero lethality (Burci et al., 2019).
Moringaceae							
174. <i>Moringa oleifera</i> Lam.	Arthritis in cattle, sheep goats (Gradé et al., 2009), cattle helminthosis (Matovu et al., 2020), NCD, diarrhoea and cough, bacterial infections and bronchitis in chicken (Zirintunda et al., 2024).	Fed directly, water extracts, added in feed formulation (Gradé et al., 2009), leaves mixed with fodder (Matovu et al., 2020), crush (Zirintunda et al., 2024)	T	L	Antimicrobial, anti-inflammatory (Henuk, 2018), Anti-inflammatory, antioxidant, antimicrobial, antileishmanic, antihepatotoxic (Singh et al., 2023).	Alkaloids, flavonoids, anthocyanins, proanthocyanidins and cinnamates (Singh et al., 2023).	
Musaceae							
175. <i>Musa</i> $\times$ <i>paradisica</i> L.	ECF in cattle (Tabuti et al., 2003a)	Prepared as a beer and given when warm (Tabuti et al., 2003a).	H	F			
Myrtaceae							
176. <i>Psidium guajava</i> L.	Coccidiosis (Bukonya-Ziraba and Kamoga, 2007), Newcastle disease (Zirintunda et al., 2024)	Decoction (Bukonya-Ziraba and Kamoga, 2007), crush (Zirintunda et al., 2024)	T	L	Antimicrobial activity (Gutiérrez et al., 2008)		
177. <i>Syzygium cumini</i> (L.) Skeels	Haematuria (Rajkumari et al., 2014), cough and diarrhoea in chicken (Zirintunda et al., 2024).	Decoction, seeds powdered (Rajkumari et al., 2014), pound leaves and bark	T	L, B	Anti-salmonella activity (Bari et al., 2024).	Glucoside jamboline, resin, tannins and phenolics (Srivastava and Chandra, 2013), betulinic acid, β -sitosterol, friedelin, epi-friedelanol (Sengupta and Das, 1965).	
Nyctaginaceae							
178. <i>Boerhavia diffusa</i> L.	Singamus (Bukonya-Ziraba and Kamoga, 2007), ECF in cattle (Tabuti et al., 2003a)	Decoction (Bukonya-Ziraba and Kamoga, 2007), infusion (Tabuti et al., 2003a).	Sh	R, Ap			
Oleaceae							
179. <i>Olea europaea</i> L.	Intestinal parasites (Gradé et al., 2009).	Water extract (Gradé et al., 2009).	T	L		Alkaloids, phenolics, terpenoids, flavonoids, and saponins, oleuropein (Adem et al., 2020).	
180. <i>Olea europaea</i> L. subsp. <i>africana</i> (Mill.) P.S. Green	Intestinal parasites (Gradé et al., 2009).	Water extract (Gradé et al., 2009).	Sh	L	Antiparasitic (Siyadatpanah et al., 2022)		
Orobanchaceae							
181. <i>Striga hermonthica</i> (Delile) Benth.	Newcastle disease in chicken (Zirintunda et al., 2024)	Crushing (Zirintunda et al., 2024)	H	Fl			
Oxalidaceae							
182. <i>Oxalis corniculata</i> L.	Influenza, enteritis, diarrhoea, traumatic injuries, sprains, and snake bites (Rashmi and Sahu, 2020), fowl typhoid, coccidiosis (Bukonya-Ziraba and Kamoga, 2007).	Infusions/decoction (Bukonya-Ziraba and Kamoga, 2007)	H	L, WP	Anti-oxidant, wound healing properties, antimicrobial (Duc et al., 2023).	Flavonoids, tannins (Duc et al., 2023).	
Phyllanthaceae							

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Table 1 (continued)

Scientific Name and (Family)	Diseases/ethnoveterinary use and animals treated	Method of preparation and use	LF	Part used	Bioactivity related to traditional uses in Ugnada	Phytochemical composition	Toxicity
183. <i>Bridelia micrantha</i> (Hochst.) Baill.	Retained placenta in cattle and sheep (Gradé et al., 2009).	Water extract which is administered orally in cattle & sheep (Gradé et al., 2009).	T	B	Anthelmintic, antimicrobial, anticonvulsant and sedative, antidiabetic, antidiarrhoeal, antinociceptive, antioxidant and β -lactamase inhibitory activities (Maroyi, 2017).	Brideliol, bridelic acid, β -sitosterol, lupeol, betulinic acid, oleanolic acid, ursolic acid, kaempferol, quercetin, rutin and catechin (Maroyi, 2017).	LD ₅₀ of the stem bark's aqueous extract in mice was 3.9 g/kg (Maroyi, 2017).
184. <i>Bridelia scleroneura</i> Müll	Retained placenta in common livestock such cattle (Gradé et al., 2009).	Water extract given in common livestock (Gradé et al., 2009).	T	B	Antibacterial, antifungal, anti-inflammatory, & analgesic (Ngueyem et al., 2009; Théophile et al., 2006)	Quercetin, kaempferol, rutin, catechin, gallic caffeic, chlorogenic, & ellagic acid (Ngueyem et al., 2009).	Low toxicity LD ₅₀ =2.5–5 g/kg (Ngueyem et al., 2009).
185. <i>Flueggea virosa</i> (Roxb. ex Willd.) Royle	Bacterial infections in chicken (Zirintunda et al., 2024)	Pounding	Sh	Rt	Antimicrobial effect (Bailly, 2024).		
Sapotaceae							
186. <i>Vitellaria paradoxa</i> C.F. Gaertn.	Constipation, diarrhoea, Mange, stomach pain in goats and camels (Gradé et al. (2009), Newcastle disease in chicken (Zirintunda et al., 2024)	Extract oil from seeds which is topically in goats and camels (Gradé et al., 2009), poud roots and give	T	Fl, Sd	Anti-inflammatory, antioxidant &, gastroprotective (Dangnon et al., 2025).	Phenolics, flavonoids, terpenoids, steroids, saponins, alkaloids, and tannins (Sinan et al., 2020).	Considered non-toxic due to its widespread use by humans and animals for various purposes (Sinan et al., 2020).
187. <i>Calotropis procera</i> (Aiton) W.T. Aiton	East coast fever, anaplasmosis in sheep & goats (Gradé et al., 2009).	Water extract administered orally in sheep and goats (Gradé et al., 2009).	H	Wp	Anti-inflammatory, analgesic, antipyretic, anthelmintic, antimicrobial (Kaur et al., 2021).	Calotropin, calactin, uscharin, proceroside, calotoxin, calotropagenin, & gigantol (Kaur et al., 2021; Wadhwani et al., 2021).	Toxic to humans and animals. Latex can cause skin irritation, blistering, inflammation and ulceration (Wadhwani et al., 2021).
188. <i>Phyllanthus</i> sp.	Snake bite (Gradé et al., 2009)	Water extract (Gradé et al., 2009)	H	Wp		Phenolics tannins, saponins, terpenoids, coumarins, and flavonoids (Nahar et al., 2011).	
Papaveraceae							
189. <i>Argemone mexicana</i> L.	Fowl typhoid (Bukonya-Ziraba and Kamoga, 2007)	Cold water extract (Bukonya-Ziraba and Kamoga, 2007).	H	L	Antibacterial activity (Saranya et al., 2012)		
190. <i>Papaver somniferum</i> L.	Heartwater (Gradé et al., 2009)	Crushed leaves nasally (Gradé et al., 2009).	H	L	Antispasmodic, analgesic (Lim, 2012).	Morphine, codeine, thebaine, papaverine, noscapine, sanguinarine, and rhoeadine alkaloids, phenols, anthocyanins, & flavonoids (Lim, 2012).	
Pedaliaceae							
191. <i>Sesamum angustifolium</i> (Oliv.) Engl.	Parasites, wounds, diarrhoea, constipation, fever and respiratory ailments in cattle., lorms, laminitis, arthritis, colic, ulcers, and anaemia, helminthes, coccidiosis and boosts immunity, increases egg production in poultry,TB (Gradé et al., 2009)	Water extract	H	R	Anti-inflammatory, antioxidant, antipyretic, analgesic, anthelmintic, antibacterial (Siamayuwa et al., 2020).	Sesamin, sesamol, sesaminol, sesamol, quercetin, kaempferol, rutin, catechin are some of the main flavonoids (Siamayuwa et al., 2020).	
192. <i>Sesamum</i> sp.	Retained placenta tuberculosis (Gradé et al., 2009)	Water extract (Gradé et al., 2009)	H	R		Lignans, phytosterols, tocopherols, phenolic acids, and flavonoids (Kamal-Eldin et al., 1994).	No negative effects 2000 mg/kg in seeds (Kamal-Eldin et al., 1994).
Plumbaginaceae							
193. <i>Plumbago zeylanica</i> L.	Contagious, bovine/caprine pleuropneumonia, Pneumonia (Gradé et al., 2009)	Water extract (Gradé et al., 2009)	Sh	B, L, R	Antibacterial, anti-inflammatory, and wound healing (Shukla et al., 2021).	Alkaloids, flavonoids, phenols, tannins, saponins, steroids, terpenoids, and naphthoquinones, plumbagin (Roy et al., 2022)	
Phytolaccaceae							
194. <i>Phytolacca dodecandra</i> L'Hér.	Schistosomiasis in cattle and goats, helminthosis in all livestock and	Extracts administered orally and topically (Gradé et al., 2009). Burn	Sh	F, L, R	Molluscicidal/ antischistosomiasis,	Saponins, tannins, flavonoids, terpenoids, phenolics, genins and	Harmful to both aquatic invertebrates and fish. High doses cause reduced

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Table 1 (continued)

Scientific Name and (Family)	Diseases/ethnoveterinary use and animals treated	Method of preparation and use	LF	Part used	Bioactivity related to traditional uses in Ugnada	Phytochemical composition	Toxicity
	poultry (Gradé et al., 2009), worms and fowl typhoid (Bukunya-Ziraba and Kamoga, 2007), retained placenta, constipation, ECF (Matovu et al., 2020)	to ash with skin of snake, make a filtrate (Bukunya-Ziraba and Kamoga, 2007), macerate/decoct and drench (Matovu et al., 2020).			antibacterial, antifungal, and anthelmintic (Beressa et al., 2020), larvicidal, insecticidal, antibacterial, antifungal, and anti-inflammatory activity (Desta and Abd El-Aty, 2021).	glycosides (Beressa et al., 2020; Desta and Abd El-Aty, 2021).	appetite, excessive urination, shivering, and sleepiness (Beressa et al., 2020).
Poaceae							
195. <i>Cymbopogon citratus</i> (DC.) Stapf	NCD (Zirintunda et al., 2024)	Crushing	H	L	Antibacterial activity (Machraoui et al., 2018).		No observed adverse effects in rats (Fandohan et al., 2008).
196. <i>Cynodon dactylon</i> (L.) Pers.	Body weakness in chicken (Zirintunda et al., 2024)	Pounding (Zirintunda et al., 2024)	H	L, St			
197. <i>Dactyloctenium aegyptium</i> (L.) Willd.	Diarrhoea (Gradé et al., 2009).	Water extract/fodder (Gradé et al., 2009).	H	Fr	Larvicidal (Muthu et al., 2025), antidiarrhoeal (Muthu et al., 2025).	Pyrimidines, tricin, phenolic acids, β -sitosterol glucoside, asyngangoside (Kayed et al., 2015; Kumar, 2025).	Cytotoxic against HeLa, A549, and HepG2 cancer cell lines (Kayed et al., 2015).
198. <i>Digitaria abyssinica</i> (Hochst. ex A. Rich.) Stapf	Helminthosis in all livestock & poultry (Gradé et al., 2009), Cattle helminthosis (Matovu et al., 2020)	Boil leaves and roots, cool then drench (Nabukunya et al., 2014)/graze animals/administer as feeds (Matovu et al., 2020).	H	L	Antimicrobial (Sapunyo et al., 2023)	Alkaloids, glycosides, phenolics, coumarins, tannins, flavonoids, steroids, and terpenoids (Sapunyo et al., 2023)	
199. <i>Hyparrhenia rufa</i> (Nees) Stapf	Retained placenta (Gradé et al., 2009)	Root decoction and drench animal (Gradé et al., 2009)	H	L			
200. <i>Sorghum bicolor</i> (L.) Moench	Lice, fleas, mites, mange and wounds (Nabukunya et al., 2014), Diarrhoea in chicken (Zirintunda et al., 2024).	Mix crushed fruits with roots of <i>C. adenocaula</i> and apply on wound-Smear on skin (Nabukunya et al., 2014). Add to feeds (Zirintunda et al., 2024).	H	Sd	Antibacterial activity (Kil et al., 2009).		
201. <i>Sporobolus pyramidalis</i> P. Beauv.	Newcastle disease in chicken (Zirintunda et al., 2024)	Pounding (Zirintunda et al., 2024)	H	R	Anti microbial (Gebashe et al., 2020)	Flavonoids, proanthocyanidins, iridoids and phenolic acids (Gebashe et al., 2020)	
202. <i>Zea mays</i> L.	Diarrhoea in chicken (Zirintunda et al., 2024)	Eaten directly	H	Sd	Antimicrobial effect (Balasundram et al., 2006)	Steroids (sitosterol and stigmasterol), saponin, tannin, alkaloid, flavonoids (Žilić et al., 2016)	
Portulacaceae							
203. <i>Portulaca psammotropa</i> Hance	Bacillary white diarrhoea (Gradé et al., 2009), coccidiosis, bacterial infections, chronic respiratory disease, infectious coryza, fowl pox, histomoniasis, Ectoparasites and cough, infectious bronchitis, fowl and Pullorum/Salmonellosis in chicken (Zirintunda et al., 2024)	Cold water extract (Gradé et al., 2009) Pound (Zirintunda et al., 2024)	H	Wp			
204. <i>Talinum afrum</i> (Thunb.) Eckl. & Zeyh.	Low milk production, eye infections, diarrhoea, wounds (Gradé et al., 2009)	Water extract (Gradé et al., 2009)	H	R			
205. <i>Talinum portulacifolium</i> (Forssk.) Asch. ex Schweinf.	Anaphylaxis (Bukunya-Ziraba and Kamoga, 2007)	Cold water extract	H	L		Steroids, terpenoids, alkaloids, glycosides, flavanoids, saponin (Rao et al., 2008).	
Primulaceae							
206. <i>Maesa latifolia</i> A.DC.	Babesiosis, aspiration pneumonia, diarrhoea, and rabies (Gradé et al., 2009).	Decoctions infusions (Gradé et al., 2009)	T	L, R	Antibacterial, anti-inflammatory (Barman et al., 2025; Bayisa and Bultum, 2022).	Flavonoids, alkaloids, tannins, saponin, Terpenoids, phenols, benzoquinones, e.g., maesinin, dihydromaesinin (Barman et al., 2025; Bayisa and Bultum, 2022).	Generally considered safe LD ₅₀ > 2 g/kg in mice (Megersa et al., 2023).

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Table 1 (continued)

Scientific Name and (Family)	Diseases/ethnoveterinary use and animals treated	Method of preparation and use	LF	Part used	Bioactivity related to traditional uses in Ugnada	Phytochemical composition	Toxicity
Proteaceae							
207. <i>Protea gaguedi</i> J.F. Gmel.	Anthrax (Gradé et al., 2009)	Water extract	T	B, L			
Rhamnaceae							
208. <i>Ziziphus mauritiana</i> Lam.	Constipation, vomiting, nausea with salivation (Gradé et al., 2009)	Water extract	T	B	Anti-inflammatory, antibacterial, immunomodulatory & antipyretic (Prakash et al., 2021).	Terpenoids, flavonoids, alkaloids, & saponins (Prakash et al., 2021).	Nontoxic at moderate doses (up to 2000 mg/kg, but could be injurious at high doses (Yahaya et al., 2019).
Rubiaceae							
209. <i>Gardenia ternifolia</i> subsp. <i>jovis-tonantis</i> (Welw.) Verdc.	Bloat, heartwater, snakebite, snake spit, stick for spinning milk (Gradé et al., 2009)	Water extract with <i>Steganotaenia</i> , bark and stem, wash eyes (Gradé et al., 2009)	T	R, B	Antibacterial, antisickling (Ngbolua et al., 2015).	Polyphenols, anthocyanins, flavonoids, tannins, coumarins, alkaloids, saponins (Ngbolua et al., 2015).	
210. <i>Gardenia ternifolia</i> Schumacher & Thonn.	ECF in cattle (Tabuti et al., 2003a)		Sh	R	Larvicidal (Ochieng et al., 2010)	Naringenin-7-O-methylether, quercetin-4, kaempferol-7-O-methylether, stigmasterol, β -sitosterol (Ochieng et al., 2010).	
211. <i>Gladiolus</i> Spp	Low milk production in cattle, sheep, goats, camels, horses, and donkeys (Gradé et al., 2009).	Water extract orally applied (Gradé et al., 2009)	H	R		Arabinogalactan-protein (Gleeson and Clarke, 1979).	
212. <i>Neonauclea media</i> (Havil.) Merr.	Diarrhoea in cattle (Tabuti et al., 2003a)	Infusion. Give calf 500 ml of once a day (Tabuti et al., 2003a).	Sh/T	R	Antidiarrhoeal (Gidado et al., 2005)	Tannins, alkaloids, flavonoids, saponins, cyanogenic glycosides, anthraquinones, coumarins, monoterpenes (Fadipe et al., 2013).	
Rutaceae							
213. <i>Clausena anisata</i> (Willd.) Hook.f. ex Benth.	Calf diarrhoea fever in cattle and goats (Gradé et al., 2009)	Paste enema (Gradé et al., 2009)	Sh	R		Carbazole alkaloids e.g., clausamine H, coumarins, porphyrin derivatives, and limonoids (Lawal, 2020), Clausamines A–E, ekeberginine, Clausine F, O-demethylmurrayanine (Ito et al., 2000).	Nontoxic to melanocytes at 15 μ g/mL and but inhibited cellular tyrosinase with IC ₅₀ of 1.56 μ g/mL (Thibane et al., 2019).
214. <i>Citrus</i> Spp	Rinderpest (Gradé et al., 2009), ECF in cattle (Tabuti et al., 2003a)	Paste enema (Gradé et al., 2009)	T	R	Insecticidal potential (Ribeiro et al., 2020).	Monoterpene and sesquiterpenes (Orthman et al., 2022)	
215. <i>Citrus × limon</i> (L.) Osbeck	ECF in cattle (Tabuti et al., 2003a)	Decoction given as a prophylactic (Tabuti et al., 2003a)	Sh/T	F	Anthelmintic (Yilmaz and Yildiz, 2023).	Alkaloids, flavonoids, steroids, terpenoids, saponins, cardiac glycosides (Oikeh et al., 2016)	
216. <i>Citrus aurantium</i> f. <i>aurantium</i>	Newcastle disease in chicken (Zirintunda et al., 2024), Coccidiosis in chicken (Zirintunda et al., 2024)	Crushing (Zirintunda et al., 2024)	T	L	Antibacterial activity (Okla et al., 2019)	4-terpineol, D-limonene, 4-carvo-menthenol & linalool (Okla et al., 2019), octopamine, neohesperidin; nobiletin, 5-odesmethyl nobiletin, quinoline; narcotin; noradrenline, tryptamine, tyramine; naringin; rhoifolin; Ionicerin (Suryawanshi, 2011)	Increased blood levels of cyclosporine, causing toxicity (Bent et al., 2004), synephrine and octopamine may cause hypertension (Nykamp et al., 2004) and cardiac arrhythmias leading to heart attack (Bui et al., 2006).
217. <i>Fagaropsis angolensis</i> (Engl.) H.M. Gardner	Anaplasmosis, contagious bovine/caprine, pleuropneumonia, lice, snake bite (Gradé et al., 2009)	Water extract topically applied (Gradé et al., 2009)	T	B	Antibacterial, antifungal, antiviral, antiprotozoal, anti-inflammatory (Mutinda et al., 2022).	Diesters, a tiglane diterpene, euphol, euphorbinol, euphorone, euphorbol, euphorginol, euphorinol, euphorbin, euphorbinone, euphorbinene (Mutinda et al., 2022).	Highly toxic to Vero cells (Gaobotse et al., 2023).
218. <i>Harrisonia abyssinica</i> Oliv.	Contagious bovine/caprine, pleuropneumonia, chronic neck/back wound with maggots, dermatophilosis, fly and maggot control, Mange, heartwater, rectal	Water extract orally applied Poultice with oil/water/urine extract and butter, topically applied (Gradé et al., 2009), infusion/decoction with above	Sh	R	Antimicrobial, antifungal, (<i>A. niger</i> , <i>P. aeruginosa</i> , <i>C. albicans</i> , and <i>S. aureus</i>) anti-inflammatory, antipyretic (Egbèyèmi et al., 2022).	Alkaloids, flavonoids, tannins, saponins, steroids, terpenoids, coumarins, quinones, harrisonine, harrabyssinol, harrabyssinone, harrabyssic acid, harrabyssolide,	Low toxicity of leaf extracts in rats (LD ₅₀ < 2000 mg/kg bw (Egbèyèmi et al., 2022).

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Table 1 (continued)

Scientific Name and (Family)	Diseases/ethnoveterinary use and animals treated	Method of preparation and use	LF	Part used	Bioactivity related to traditional uses in Ugnada	Phytochemical composition	Toxicity
219. <i>Zanthoxylum chalybeum</i> Engl.	wound (Gradé et al., 2009), ECF and diarrhoea in cattle (Tabuti et al., 2003a), bacterial infections in chicken (Zirintunda et al., 2024). Anaplasmosis, calf diarrhoea, fever, liver disease in cattle, goats and sheep (Gradé et al., 2009), diarrhoea in chicken (Zirintunda et al., 2024)	ground parts of <i>O. sinuatum</i> /Infusion with roots of <i>V. amygdalina</i> and 'kalimi kambwa' with salt (Tabuti et al., 2003a). Water extract (Gradé et al., 2009), pounf bark (Zirintunda et al., 2024)	T	R	Antibacterial, antifungal, analgesic (Okagu et al., 2021).	harrabyssinolide (Madivoli et al., 2018). 6-benzo(1,3)dioxol-5-yl-hexa-2,5-dienoic acid isobutylamide, 4-methoxy-N-(2-methoxy-phenyl), phenyl acrylamidefagaramide (Zondegoumba et al., 2019)	Highly potent toxicity in Brine shrimp lethality assay (Mguni et al., 2023)
220. <i>Zanthoxylum leprieurii</i> Guill. & Perr.	Newcastle disease in chicken (Zirintunda et al., 2024)	Pounding (Zirintunda et al., 2024)	T	B	Antibacterial activity (Oloya et al., 2023)	1-Hydroxy-3-methoxy-N-methylacridone, scoparone, arborinine (Zondegoumba et al., 2019)	Some of its compounds are more toxic on Chang cells than to HepG2 cells (Zondegoumba et al., 2019).
221. <i>Vepris nobilis</i> (Delile) Mziray	Rheumatism, arthritis, pneumonia, cough, fever, joint anthelmintic in cattle, goats (Gradé et al., 2009)	Water extracts, decoction (Gradé et al., 2009)	Sh	L, St, B, R, Fr,	Antipyretic, analgesic, anti-inflammatory, antimicrobial, antileishmanial and antitrypanosomal (Omujal et al., 2020).	Quinoline and furoquinoline alkaloids, terpenoids and flavonoids (Omujal et al., 2020).	Immunotoxicity and mutagenicity (Chepkirui and Kagia, 2020)
Salicaceae							
222. <i>Oncoba spinosa</i> Forssk.	ECF in cattle (Tabuti et al., 2003a)	Infusion/dried fruits tied around neck of calf to act as a prophylactic (Tabuti et al., 2003a).	Sh	R, F	Anti-parasitic activity (Kodi, 2018)	Kaempferol, quercetin, apigenin-7-O- β -D-glucuronopyranoside, quercetin 3-O- β -D-galactopyranoside (Djouossi et al., 2015).	
Solanaceae							
223. <i>Capsicum annum</i> L.	Contagious bovine/caprine, pleuropneumonia, heartwater in livestock (Gradé et al., 2009), Coccidiosis in chicken (Zirintunda et al., 2024), NCD (Nabukanya et al., 2014), bacillary white diarrhoea, coccidiosis, fowl typhoid (Bukenya-Ziraba and Kamoga, 2007).	Water extract with <i>W. Warburgia</i> spp bark, <i>S. incanum</i> fruits, sediment rock (CaCO ₃) and water/ Paste, nasally/pound fruit and apply to freshly branded areas (Gradé et al., 2009), Add ash and water to freshly squeezed leaves then drench/mix with <i>C. sativa</i> , <i>A. sisalana</i> juice and ash then drench/Mix with <i>Aloe</i> spp (Nabukanya et al., 2014)	H	F, F	Anthelmintic, insecticidal and insect repellent, antimicrobial activity (Sann, 2020).	Capsaicinoids, carotenoids, flavonoids, phenolics, alkaloids, saponins, tannins, phlobatannins, anthraquinones and cardiac glycosides (Wahua et al., 2013)	Safe for human & animal consumption. In a 4-week feeding study, <i>C. annum</i> in the diet at concentrations up to 10 % was relatively nontoxic in groups of male mice (Wilbur, 2007).
224. <i>Capsicum frutescens</i> L.	Diarrhoea in chicken (Zirintunda et al., 2024)	Crush (Zirintunda et al., 2024).	H	L	Antibacterial activity (Emeka et al., 2022)	Tropane alkaloids (El Bazaoui et al., 2011)	No significant gross pathology in rats, but slight hyperemia of the liver and reddening of the gastric mucosa (Wilbur, 2007).
225. <i>Nicandra physalodes</i> (L.) Gaertn.	Cattle helminthosis (Matovu et al., 2020), ECF in cattle (Tabuti et al., 2003a)	Pound 2 handfuls add water/ macerate and drench (Matovu et al., 2020), mixed with milk of the mother cow (Tabuti et al., 2003a).	H	L	Insecticidal property (Gupta et al., 2014)	Alkaloids like withanolide, withaperuvine E and nicandrin B (Bagchi et al., 1984)	Clinical signs of poisoning appear within 6 h and include tachycardia, circling, tremors of the hind limbs, bloat, convulsions, coma and death (Gupta et al., 2014)
226. <i>Nicotiana tobacum</i> L.	Fleas (Gradé et al., 2009), ticks (Byaruhanga et al., 2015), cattle helminthiasis (Matovu et al., 2020), NCD, ectoparasites in chicken (Zirintunda et al., 2024)	Urine extract, add ash and wash/ crush fresh leaves, mix with water and wash (Byaruhanga et al., 2015; Gradé et al., 2009), pound to make paste/macerate & squeeze 4-5drops in eye (Matovu et al., 2020).	H	L	Antiparasitic effect (Sharma et al., 2016)	Saponin, flavonoids and alkaloids (Sharma et al., 2016)	Neuronal and liver degeneration, necrosis and cirrhosis (Shekins et al., 2016).
227. <i>Solanum aculeastrum</i> Dunal	Retained placenta	Squeeze fruits in water/milk and drench (cough)/heat ripe fruits and	Sh	Fl		Alkaloids (Wanyonyi et al., 2002), tannins and terpenoids (Hikaambo et al., 2023)	

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Table 1 (continued)

Scientific Name and (Family)	Diseases/ethnoveterinary use and animals treated	Method of preparation and use	LF	Part used	Bioactivity related to traditional uses in Ugnada	Phytochemical composition	Toxicity
228. <i>Solanum aculeatissimum</i> Jacq.	Anaplasmosis (Gradé et al., 2009).	smear on vulva (Nabukenya et al., 2014; Zirintunda et al., 2024). Water extract (Gradé et al., 2009).	Sh	Fl	Antibacterial (Koduru et al., 2006).	Alkaloids, flavonoids, phenols, saponins, steroids, and terpenoids (Nabeta, 1993).	No toxic in rats (Kunwar et al., 2020)
229. <i>Solanum dasyphyllum</i> Schumach. & Thonn.	Contagious bovine/caprine Pleuropneumonia, Heartwater, Poultry respiratory infection(Gradé et al., 2009).	Water extract	H	F	Antibacterial activity (Belitibo et al., 2024)	Saponins, tannins, alkaloids, flavonoids, steroids and cardiac glycosides (Matthew et al., 2021)	Toxic glycoalkaloids which cause diarrhoea/carcinogenic glycosides (Eletta et al., 2017)
230. <i>Solanum incanum</i> L.	Anthrax, inappetence, diarrhoea, respiratory diseases and tick infestations, abscess (Gradé et al., 2009), retained placenta (Nabukenya et al., 2014), ECF (Matovu et al., 2020), cataract in cattle (Tabuti et al., 2003a), wounds on eye in cattle (Tabuti et al., 2003a; Zirintunda et al., 2024).	Pound (Gradé et al., 2009), smear crushed fresh fruit on vulva (Nabukenya et al., 2014), decoct/macerate and drench (Matovu et al., 2020). Use sap/mix sap with powder from burnt snail shell/salt/cassava flour and apply to eye (Tabuti et al., 2003a)/ash from leaves and fruits (Zirintunda et al., 2024).	H	L, St, Fr	Antimicrobial, antiparasitic (Sbhatu and Abraha, 2020).	Steroids, flavonoids, quercetin, solasodine, β -Sitosterol, stigmaterol, solamargine, yamogenin (Rani et al., 2025).	No clear acute toxicity of fruits crude extracts in wiss albino mice, but showed toxic effects and mortality at high doses (400 mg/kg) (Feyera et al., 2017)
231. <i>Solanum giganteum</i> Jacq.	Pain (Gradé et al., 2009).	Flesh around seeds consumed direct (Gradé et al., 2009).	H	F		Alkaloids (Chowański et al., 2016)	
232. <i>Solanum</i> sp.	Otitis with pus, often ticks, Anaplasmosis, East coast fever, heartwater, liver flukes, rinderpest	Water extract with sediment rock (CaCO ₃) and <i>Warburgia Warburgia</i> spp bark	H	F	Insecticidal activity against various insect species (Chowański et al., 2016)	Polyphenols, glycoalkaloids and saponins (Yuan et al., 2020)	Steroidal glycoalkaloids toxic to many organisms (Akiyama et al., 2023)
233. <i>Withania somnifera</i> (L.) Dunal	Improve broiler performance and carcass attributes in poultry (Gradé et al., 2009).	Water extract(Gradé et al., 2009).	H	Wp		Steroidal lactones called withanolides, phenolic acids, flavonoids, and alkaloids (Polumackanycz et al., 2023).	Cardiac toxicity in adult Wistar rats (Mahdi and Al-Okialy, 2019)
Theaceae							
234. <i>Camellia sinensis</i> (L.) Kuntze	Newcastle disease in chicken & Fowl cholera (Zirintunda et al., 2024).	Crush leaves (Zirintunda et al., 2024)	T	L	Antibacterial activity (Al-Aboudi and AL-Azawi, 2024)	Phenolic compounds (Camargo et al., 2016)	
Urticaceae							
235. <i>Laportea aestuans</i> (L.) Chew	Diarrhoea, dysentery, fevers, ulcers, oedema (Gradé et al., 2009).	Water extract, dried powder orally, topically as poultice (Gradé et al., 2009).	H	L, St, Fl	Anti-helminthic (Oloyede and Ayanbadejo, 2014).	Saponins, alkaloids, cardiac glycosides, phenolics, tannins, flavonoids (Emeka et al., 2022)	Genotoxic (Emeka et al., 2022).
Verbenaceae							
236. <i>Lantana camara</i> L.	Measles in cattle (Tabuti et al., 2003a)	Infusion made after adding leaves of <i>C. opulifolium</i> and <i>C. sativa</i> (Tabuti et al., 2003a).	Sh	L	Antiviral activity (García et al., 2010).	Tannins, steroids, flavonoids, quinones, phenols, caumarins, alkaloids (Mansoori et al., 2020), lantadenes (Sharma et al., 1991)	Poising in cattle mainly affecting the liver and kidneys the most (Sharma et al., 1981).
Vitaceae							
237. <i>Cissus cactiformis</i> Gilg	Bloat, fractures, wound healing in cattle, sheep, goats, & dogs (Gradé et al., 2009).	Water extract which is orally applied	Sh	St	Anti-inflammatory, analgesic, antibacterial (Panthong et al., 2007).	Flavonoids, triterpenoids, phenolics, alkaloids, & steroids. β -cholest-3-ene (Yoon et al., 2023)	
238. <i>Cissus quadrangularis</i> L.	Anaplasmosis, contagious bovine/caprine, pleuropneumonia, calf diarrhoea, intestinal parasites, low milk production (Gradé et al., 2009).	Water extract for infections, and wounds, and sheep, goats, horses, and dogs (Gradé et al., 2009).	Sh	L	Antibacterial (Anwar et al., 2020)	Flavonoids, triterpenoids, phenolics, alkaloids, steroids, β -sitosterol, kaempferol, resveratrol, β -amyirin, α -amyirin, & quercetin (Bafna et al., 2021)	
239. <i>Cissus rotundifolia</i> Lam.	Rotten wounds, mouth wounds of cows, goats and sheep.	Water extract orally applied to wash mouth/poultice with oil or water used	Sh	Wp, L	Antibacterial, analgesic, anti-inflammatory (Bawakid et al., 2025; Shanmugam et al., 2025).	β -cholest-3-ene, heptadecanoic acid, and β -sitosterol acetate (Bawakid et al., 2025).	

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Table 1 (continued)

Scientific Name and (Family)	Diseases/ethnoveterinary use and animals treated	Method of preparation and use	LF	Part used	Bioactivity related to traditional uses in Ugnada	Phytochemical composition	Toxicity
240. <i>Cyphostemma adenocaula</i> (Steud. ex A. Rich.) Desc. ex Wild & R.B. Drumm.	Bovine Ephemeral Fever	Crush leaves, add ash and drench (Nabukenya et al., 2014)	H	L	Antimicrobial activity (Hamill et al., 2003)	Betulin, betulinic, cyphostemmic A, B, C, D, & epigouanic acids A, lupeol, zizyberanal acid, β -sitosterol (Chouna et al., 2016).	
241. <i>Cyphostemma ukereuwense</i> (Gilg) Desc.	Abscess with maggots, fly repellent, intestinal parasites, ticks (Gradé et al., 2009),	Powder topically applied, water extract/poultice (Gradé et al., 2009).	H	L			
Zingiberaceae							
242. <i>Curcuma longa</i> L.	Gastrointestinal issues, inflammation (Gradé et al., 2009), NCD in chicken (Zirintunda et al., 2024)	Powders/infusions given topically/ inhaled (Gradé et al., 2009), crush and administer (Zirintunda et al., 2024).	H	Rh	Antimicrobial, anti-oxidant, anti-inflammatory (Jyotirmayee and Mahalik, 2022).	Flavonoids, alkaloids, tannins, and saponins (Jyotirmayee and Mahalik, 2022).	Turmeric and curcumin are also safe in animals (Soleimani et al., 2018).
243. <i>Zingiber officinale</i> Roscoe	NCD and diarrhoea in chicken (Zirintunda et al., 2024).	Crush and administer (Zirintunda et al., 2024)	H	Rh	Antimicrobial, antidiarrhoeal (Shah and Krishnamurthy, 2013),	Zingerone, shogaols and gingerols, and volatile oils (Shah and Krishnamurthy, 2013).	No maternal toxicity in rats (Wilkinson, 2000).
Zygophyllaceae							
244. <i>Balanites aegyptiaca</i> (L.) Delile	Blindness, eye diseases, diarrhoea, painful red eyes, mosquito control (Gradé et al., 2009), helminthosis (Byaruhanga et al., 2015), ECF, measles, worms in cattle (Tabuti et al., 2003a).	Put sap in eye/water extract/ powder fruits mixed with ash of <i>K. africana</i> (Gradé et al., 2009). Crushed with water for drenching (Byaruhanga et al., 2015), infusion (Tabuti et al., 2003a).	T	Ex, B, R Fl L	Antibacterial, antifungal, antiprotozoal, antibacterial (Singh et al., 2017).	Flavonoids and alkaloids, Saponins - diosgenin, yamogenin and balanitin-6 (Gautam et al., 2025; Singh et al., 2017).	Generally safe for livestock, but high quantities or prolonged use may have harmful effects (Gautam et al., 2025).
245. <i>Balanites rotundifolia</i> var. <i>rotundifolia</i>	Otitis with pus, often ticks	Paste is applied in ear	T	L	Antimicrobial, (Seid et al., 2025).	Balanitoside, diosgenin, balanitin-6, spinasterol, stigmasterol, β -sitosterol, lupeol, betulin, betulinic, ursolic oleanolic acids, quercetin, alkaloids, steroids, glycosides, terpenoids, coumarins, phenols (Seid et al., 2025; Thakkar et al., 2025).	

Key: B = bark, LF = Life Form Wp = Whole plant, Fl = flowers, St = Stem, Sd = seeds, R = roots, L = leaves, Ex = Excudate

comprehensive account of the toxicity profiles of the individual plant species used. As a result, only studies which provided comprehensive information on toxicity were included in this review.

4. Plant families, life forms and parts of plants used in ethnoveterinary medicine

Twenty nine families were represented by only one plant species and 18 species were represented by two plant species each (Fig. 1). Most of the plant species used were from the Fabaceae family (38) followed by Asteraceae (16) and Euphorbiaceae (13). These were followed by Malvaceae, Rutaceae and Curcubitaceae with 9 species each.

The genus *Euphorbia* produces caustic lattices, which constitute a health hazard to humans and livestock. Direct contact of the irritant latex with the eye can cause blindness (Upadhyay et al., 1976). In this review, 13 species of the family Euphorbiaceae were cited viz *Alchornea cordifolia*, *Erythrococca bongensis*, *Euphorbia bongensis*, *Euphorbia candellabrum*, *Euphorbia cooperi*, *Euphorbia hirta*, *Euphorbia heterophylla*, *Euphorbia umbellata*, *Euphorbia tirucalli*, *Jatropha curcas*, *Acalypha fruticosa*, *Tragia brevipes*, *Ricinus communis* out of which 7 species were from the genus *Euphorbia*. *Euphorbia heterophylla* and *Euphorbia hirta* are commonly found in the Nigerian pasture and have been shown to be toxic in albino rats (Adedapo et al., 2004). Members of the Euphorbiaceae family tend to be rich in diterpenes, such as most of phorbol and ingenol esters can be considered exclusively as toxins without any possible medicinal use, whereas the other group of diterpenes consists of compounds, which display toxicity, but in adequate dose they have therapeutic perspective (e.g. the resiniferatoxin with capsaicine-like effect), and are thus promising lead compounds for natural product based drug developments (Hohmann and Molnár, 2004).

Most of the plant species used in ethnoveterinary medicine were herbs (48.4 %), followed by shrubs (26.8 %) and trees (24.8 %) (Fig. 2).

The commonest plant parts used in ethnoveterinary medicine were leaves (34.2 %), followed by roots (20.7 %) and bark (14.2 %) (Fig. 3).

5. A comparative analysis of ethnoveterinary medicine use in Uganda and the rest of Africa

The use for plants in ethnoveterinary medicine is widespread across the African continent and many farmers rely on traditional healthcare practices to keep their animals healthy and to treat them when they are sick. Ethnoveterinary practices play significantly greater roles in livestock health care as an alternative or integral part of modern veterinary practices (Dzoyem et al., 2020). A number of studies have reported the use plant species in EVM. For instance (Saganuwan, 2017), reported 215 plant species in Nigeria and (Dzoyem et al., 2020) recorded 138 plants belonging to 69 families used to manage livestock diseases in Cameroon, and 645 plant species (from 133 families) from Ethiopia (Tilahun et al., 2019; Chaachouay et al., 2022) documented 150 plant species from 129 genera and 56 families used in ethno-veterinary medicine in Morocco.

Across southern Africa, especially in South Africa, EVM is characterized by broader species use, greater phytochemical diversity, and more extensive preliminary validation. For instance (Selogatwe et al., 2021), identified 104 woody plant species from 44 plant families that are used in the treatment of different diseases in animals, particularly cattle and goats in a review of ethnoveterinary knowledge, biological activities and secondary metabolites of medicinal woody plants used for managing animal health in South Africa (McGaw et al., 2020), reported the use of 139 plants from 50 families used against 21 animal diseases and conditions in South Africa from 2009 until 2019, whereas (Gobvu et al., 2023), conducted a review of medicinal plants used for ethnoveterinary medicine, specifically for poultry in Zimbabwe and reported the use of 53 plant species among commercial poultry farmers and Chakale and colleagues evaluated the ethnoveterinary Knowledge of plants used for treating diseases of cattle in South Africa. They reported a total of 310 medicinal plants from 81 families used to treat 10

categories of cattle diseases from 7 provinces in South Africa (Chakale et al., 2021). The most dominant plant families reported in this review: Fabaceae, Asteraceae, Euphorbiaceae, Malvaceae, Rutaceae and Poaceae are similar to those reported in other reviews such as the review of ethnoveterinary medicine in various other studies (Chaachouay et al., 2022; Gobvu et al., 2023; McGaw et al., 2020; Selogatwe et al., 2021).

Chakale et al. (2021) reported a relatively low proportion (21 %) to have been evaluated for various biological activities using relevant bioassays related to cattle diseases. Similarly, few plant species in Uganda have been evaluated for various biological activities using relevant bioassays related to cattle diseases and antimicrobial activity generally is the most widely studied activity. Perhaps this could be attributed to the large number of diseases caused by microbes. This highlights the dearth of scientific research on the biological activities and efficacy of medicinal plants used in ethnoveterinary medicine. The contrast between Uganda and other African regions underscores divergent trajectories. Uganda's EVM remains primarily documented at the grassroots level, with rich ethnobotanical detail but limited laboratory validation. In comparison, southern Africa demonstrates a more holistic approach: widespread ethnobotanical documentation, early-stage scientific validation, and structured analytical methods.

6. Diseases and conditions treated using ethnoveterinary medicine

We grouped the diseases or conditions treated into 84 broad categories. However, some of the diseases/conditions treated have been listed as symptoms of some diseases. For instance, the World Health organization for animal health describes Lumpy skin disease as a viral disease of cattle characterised by fever, nodules on the skin, mucous membranes and internal organs, emaciation, enlarged lymph nodes, oedema of the skin, and sometimes death. Some of these signs and symptoms are listed as different diseases or conditions treated in the different studies cited (World Health Organisation for Animal Health, 2025). Forty nine diseases or conditions were mentioned in at least 3 different studies, while 19 were mentioned at least 10 times. The World Organisation for Animal Health produced a list (list A) of diseases with high impacts on the economy, trade, and food security. These diseases include anthrax, brucellosis, bluetongue virus, and foot and mouth disease (Clemmons et al., 2021; Penrith and Thomson, 2004).

Some of the common diseases treated are shown in Fig. 4. The most common diseases constituting about half of all the mentions were diarrhoea, helminthosis/worms (12 %), theileriosis/East Coast fever and pleuropneumonia/pneumonia at 7 %, Newcastle disease, wounds, and anaplasmosis at 6 %, coccidiosis, retained placenta, eye diseases, and ectoparasites at 5 %. These were very similar to some of the commonest diseases treated in south Africa (McGaw et al., 2020). A recent study indicated that hemoparasites including East Coast fever, anaplasmosis, coccidiosis and endoparasites such as helminths to be among the most reported diseases and conditions for which diagnostic services were needed in Uganda (Vudriko et al., 2021). Other conditions treated that were not necessarily diseases included inappetence, snakebites, injuries, sprains, low milk production and infertility.

7. Method of preparation and use of ethnoveterinary medicine

Whereas the methods of preparation and administration of ethnoveterinary medicines reported in this review are similar to those across the continent, ranging from decoctions, infusions, and topical applications, there are some unique methods (Chavula et al., 2024). reported similar methods in a review of ethnoveterinary medicine practices from Malawi, Rwanda, and Ethiopia, some practices such as mixing the medicine with milk of the mother cow in the case of *Nicandra physalodes* was reported by (Tabuti et al., 2003a), or with cow ghee or salt burnt snail shells, salt, cassava flour and urine extract. *Solanum aculeastrum* was also prepared by squeezing the fruits in water or milk and drench

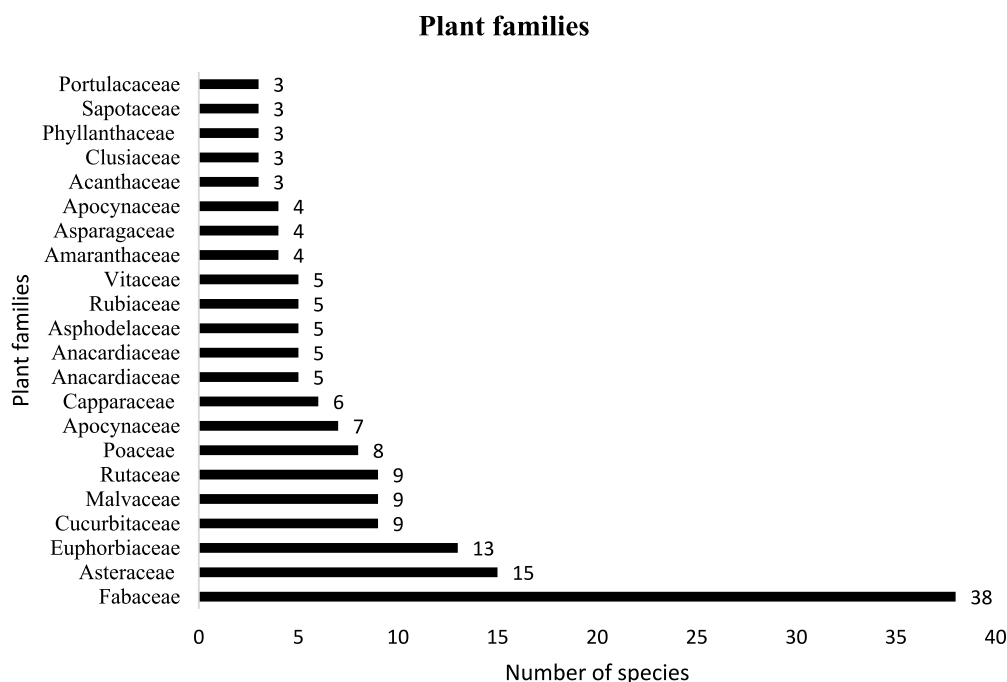


Fig. 1. Families of plant species used in ethnoveterinary medicine in Uganda.

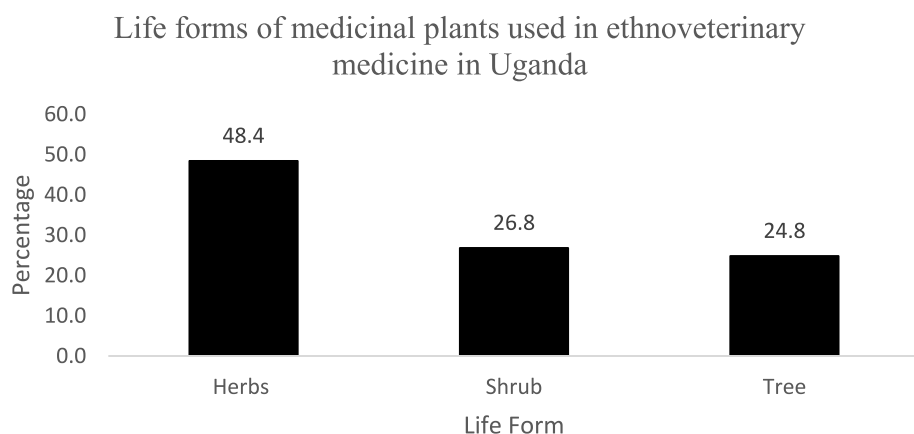


Fig. 2. Life forms of plant species used in ethnoveterinary medicine in Uganda.

cows with cough (Nabukenya et al., 2014; Zirintunda et al., 2024). Like in all other reviews considered, the predominant route of drug administration is oral, for instance in Cameroon (Dzoyem et al., 2020), Nigeria (Saganuwan, 2017), South Africa (McGaw et al., 2020) and Ethiopia (Tilahun et al., 2019) and Morocco (Chaachouay et al., 2022).

The dosages used for the herbal medicines are not precise and vary with the person treating, the condition being treated. Similar reports have been made with EVM in Morocco (Chaachouay et al., 2022). The ethnoveterinary medicines were mostly freshly prepared as and when need arose. This is a common trend in ethnoveterinary medicine use in Africa and has been reported in various studies (Eshetu et al., 2015; Graté et al., 2009).

The preparation and use of EVM are highly dependent on oral tradition and experiential learning, often restricted to elder community members or traditional healers. This knowledge is increasingly under threat due to generational gaps, urbanization, and lack of formal documentation. Furthermore, issues of standardization, dosage accuracy, and potential toxicity underscore the need for scientific validation

and integration with conventional veterinary medicine (Chaachouay et al., 2022; Tabuti et al., 2003b).

8. Bioactivity, phytochemical composition and toxicity of plants used in ethnoveterinary medicine in Uganda

Some of the medicinal plant species reported in this review have some literature reporting or confirming some aspects of their bioactivity, especially for concerning their antimicrobial activity. An extensive review of ethnoveterinary medicine use in South Africa also reported that antibacterial activity was the most studied biological activity (Chakale et al., 2021). There is also a wealth of information on the general phytochemical composition of some of the plant species, although non-specific. The phytochemical analyses were generally of a general (qualitative nature) without any specific attribution to any reported bioactivity. Also from the review, most of the plant species have not been evaluated for all their ethnoveterinary uses as well as toxicity.

Ingesting certain plants can be toxic to both animals and humans,

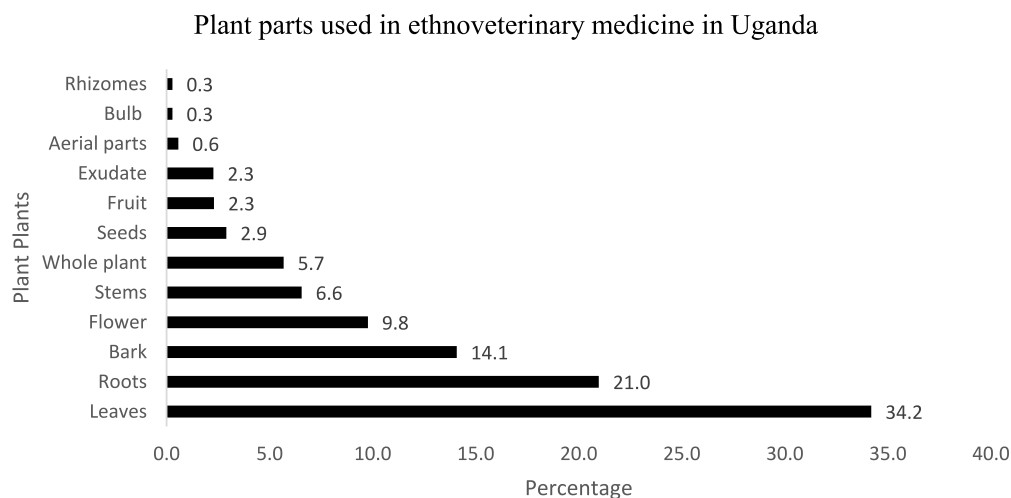


Fig. 3. Plant parts used in ethnoveterinary medicine.

potentially causing poisoning even in small amounts. This poses serious threats to livestock well-being and may hinder economic development. Incidents of livestock poisoning from medicinal and toxic plants frequently occur due to incorrect usage and the lack of established dosage standards (Abdisa and Dilbato Dinbiso, 2024; Stegelmeier et al., 2025). Poisonous plants produce secondary compounds that function as protective agents, deterring herbivores and other potential threats (Kumar et al., 2025). Livestock that graze in grasslands and forested areas are especially vulnerable to ingesting toxic plants. Key harmful secondary metabolites that pose risks to animal health and economic productivity include alkaloids found in plant families such as Asteraceae, Convolvulaceae, Lamiaceae, Fabaceae, and Boraginaceae; cyanogenic compounds present in *Sorghum* sp and other Poaceae members; nitrates from *Amaranthus* spp, *Solanum* spp, and *Chenopodium* spp; oxalates from the Poaceae, Amaranthaceae, and Polygonaceae

families; as well as various glycosides (Abdisa and Dilbato Dinbiso, 2024; Divekar et al., 2022).

The most commonly reported effects of toxic plants on livestock health in the literature include teratogenic and abortifacient, hepatotoxicity, photosensitization and impairing respiratory and circulatory systems (Abdisa and Dilbato Dinbiso, 2024; Poppenga, 2010; Stegelmeier et al., 2020). Toxic plants cause significant economic damage, both directly and indirectly. Direct impacts include livestock fatalities, miscarriages, lower milk quality, and diminished skin and hide yields, while indirect impacts involve the expenses related to treating and managing the affected animals (Keeler et al., 2013; Panter et al., 2013). Therefore, it is essential to focus on identifying the most common toxic plants, isolating their secondary metabolites, performing toxicity assessments, establishing standardized dosages, and formulating effective approaches for managing both the plants and their toxic effects (Abdisa and Dilbato

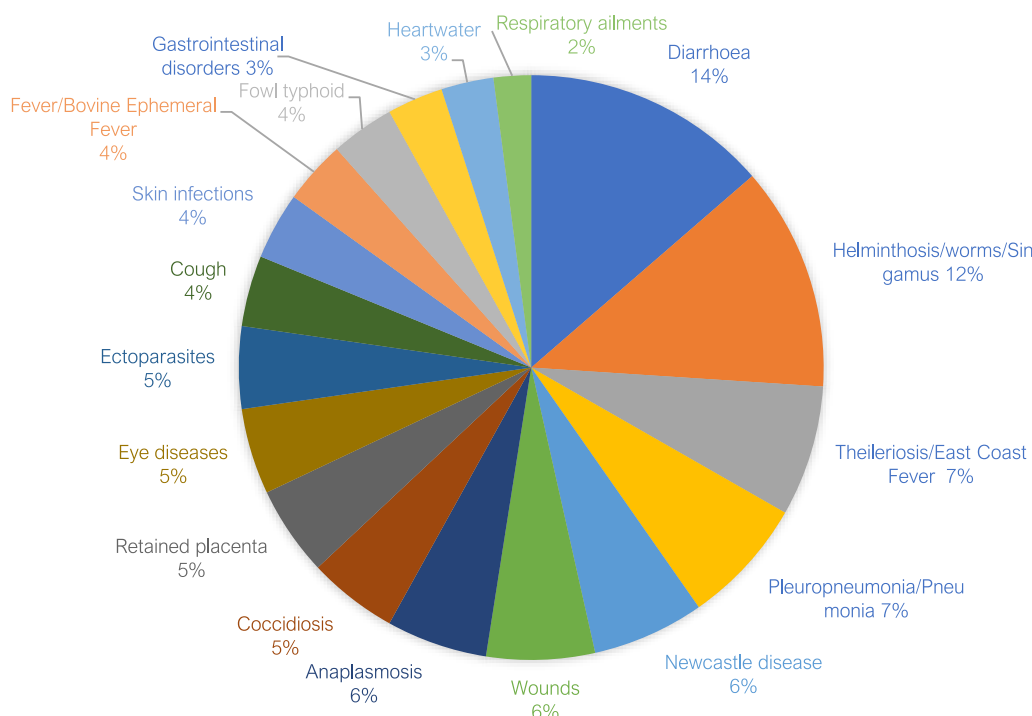


Fig. 4. Commonest disease treated.

Dinbiso, 2024).

Information on the toxicity of especially plants that are already known to be highly toxic such as *Lantana camara* and *Sarcostemma viminale* which are toxic to sheep (Jackson and Shone, 1958), and resulting in beak patting, muscular tremors and head flopping and death in ostriches (Cooper, 2007), *Ricinus communis* which contains biologically active protein ricin which has A- and B-chains that cause cell death (Bozza et al., 2015), *Senna occidentalis* which causes acute atrophy of the liver and muscles leading to death (Delachave and De Pinho, 2003), *Melia azedarach* whose poisoning may result in gastrointestinal, cardiovascular, respiratory, /neurological effects, and death in severe cases (Phua et al., 2008), plus several other examples.

The use of toxic plants in ethnoveterinary medicine has far reaching implications to both animal and human health and needs to be investigated further. Despite these challenges, ethnoveterinary medicines have the potential to foster ethnoveterinary research among livestock disease and could lead to the development of new, efficacious and safe ethnoveterinary drugs (Eiki et al., 2021).

Emerging infectious illnesses have drawn more attention in the last few decades because of the possibility that they could start pandemics and destroy the livestock industry (Jones et al., 2008). Multiple examples include the avian influenza pandemics that have decimated chicken populations worldwide and the African swine fever outbreaks that are endangering the production of pork in different parts of the world (Singh et al., 2019; Costard et al., 2013). These outbreaks highlight the need for cooperative efforts in disease surveillance, prevention, and control by demonstrating the interdependence of human livelihoods, animal health, and international trade (FAO, 2011). In addition, zoonotic diseases that people can contract from animals pose a serious threat to public health in addition to having an adverse effect on animal welfare through direct transmission or ingestion of contaminated animal products (WHO, 2020).

Gaps in our knowledge of animal diseases continue to exist despite improvements in disease surveillance and management strategies, especially in resource-constrained environments where veterinary services and surveillance equipment may be scarce (Grace et al., 2012). Furthermore, the development of alternate treatment techniques and cautious antibiotic usage practices are important in light of the impending threat that antimicrobial resistance poses to disease management (Gakuubi and Wanzala, 2012; O'Neill, 2016; Rostami et al., 2017).

Despite its historical and cultural value, ethnoveterinary medicine is relevant because it presents viable answers to current problems in animal healthcare. There is increased interest in investigating the effectiveness of ethnoveterinary techniques and incorporating them into mainstream veterinary care as traditional veterinary therapies encounter obstacles such cost, accessibility, and antimicrobial resistance (Mathias, 2004). However, EVM is frequently disregarded or ignored in favour of traditional biomedical treatments, despite its potential benefits. This neglect jeopardizes the resilience of communities and ecosystems that depend on traditional veterinary knowledge, in addition to ignoring centuries' worth of accumulated wisdom (McCorkle and Mathias-Mundy, 1992).

9. Limitations of the study

9.1. Potential biases in ethnobotanical data collection

Ethnobotanical data collection in ethnoveterinary studies is often influenced by several methodological and contextual biases that can affect the reliability and representativeness of findings. One common issue is reliance on limited and localized studies, which may capture only a narrow segment of traditional knowledge (Tabuti et al., 2010). Many investigations in Uganda focus on specific regions or ethnic groups, such as pastoralist communities traditional medicine practitioners. This makes it difficult to generalize findings across the country's

diverse cultural and ecological zones (Nabatanzi et al., 2025).

Selection bias is another concern, as researchers often depend on easily accessible or well-known healers, potentially overlooking marginalized or less visible practitioners whose knowledge might differ substantially (Martin, 2010). Likewise, recall bias can occur when informants describe plant uses based on memory rather than observation, leading to incomplete or exaggerated reports (Cunningham, 2001).

Regional variations in plant availability and disease prevalence influence reported uses. Species applied in the drylands of Karamoja may differ entirely from those used in humid highland areas (Gradé et al., 2009). Such ecological variability, combined with linguistic and cultural diversity, complicates comparative analysis across districts and studies. Publication bias may also arise because many ethnobotanical surveys remain unpublished or appear in local grey literature, while studies reporting novel species or bioactive compounds are more likely to be disseminated (Hamilton, 2004). This can also be attributed to the limited research funding into this area. Addressing these biases requires methodological triangulation, inclusive sampling, and standardized documentation protocols to ensure that Uganda's rich ethnoveterinary knowledge is accurately represented and preserved.

9.2. Critical appraisal of the quality of included studies

While numerous studies have documented the ethnoveterinary use and biological activity of medicinal plants in Uganda, a closer examination reveals considerable methodological variability that limits the strength and comparability of findings. Many bioactivity assays are preliminary screenings, often conducted using crude extracts with poorly defined extraction procedures, solvent choices, and concentrations. This lack of standardization makes it difficult to attribute observed effects to specific phytochemical constituents or to reproduce results across laboratories.

Moreover, phytochemical analyses in several reports remain largely qualitative. Such limited characterization constrains understanding of dose-response relationships and bioactive compound mechanisms. In bioactivity evaluations, methodological flaws include inappropriate model selection, small sample sizes, and insufficient replication. Some antimicrobial assays employ outdated diffusion methods lacking positive controls or reference antibiotics, which can overestimate potency (Benkova et al., 2020). Similarly, some antioxidant assays are rarely validated with standardized protocols or *in vivo* models, leading to exaggerated claims of efficacy (Gulcin, 2025). Many studies infer safety from traditional use without conducting acute or sub-chronic toxicity tests. This poses a risk for translating EVM formulations into formal veterinary practice.

This review was constrained by the scarcity of comprehensive ethnobotanical, bioactivity and phytochemical data on Ugandan plant species, particularly grey literature and unpublished ethnoveterinary local knowledge. The few studies analysed largely focus on particular animal species especially poultry or specific conditions such as tick management. There is also no data for several regions in the country, for instance, some of the areas in the cattle corridor, especially in western Uganda. This makes it difficult to have a comprehensive assessment of ethnoveterinary knowledge in Uganda.

We recommend more detailed and comprehensive ethnoveterinary studies in Uganda, by multidisciplinary research teams. We also recommend toxicological profiling of medicinal plants to come up with a regional atlas of the most troublesome plant but also useful plants for animal diseases based on scientific evidence. We also recommend exploration of how EVM can be best integrated with modern veterinary systems and establishment of open-access digital databases specific to Ugandan ethnoveterinary medicinal plants, documenting uses, bioactivity, phytochemistry, and known toxicity profiles.

10. Integration of ethnoveterinary medicine into national disease control programmes in Uganda

Ethnoveterinary medicine (EVM) can play a vital complementary role in Uganda's national disease control efforts by enhancing community participation, surveillance, and treatment accessibility. Traditional livestock keepers and healers possess long-standing diagnostic knowledge that can strengthen community-based disease surveillance, particularly in remote pastoral areas like Karamoja, where formal veterinary coverage is limited (Gradé et al., 2009). Their early recognition of disease syndromes can improve outbreak detection and response within the "One Health" framework (Buregyeya et al., 2020).

Validated herbal remedies offer affordable, locally available treatments that can supplement conventional pharmaceuticals, especially during shortages (Patwardhan and Partwardhan, 2005). Integrating such remedies into district veterinary formularies after safety and efficacy validation would enhance accessibility and sustainability. Collaborative training and extension programmes can bridge traditional and formal systems, improving knowledge exchange, sustainable harvesting, and standardization of herbal practices (Nalule et al., 2011). Finally, incorporating EVM within national policy frameworks under the Ministry of Agriculture, Animal Industry and Fisheries (MAAIF) would ensure recognition, research funding, and equitable benefit sharing. Together, these strategies support culturally grounded, cost-effective, and sustainable livestock health systems that complement conventional disease control programmes in Uganda.

11. Conclusion

Ethnoveterinary medicine (EVM) remains a cornerstone of animal healthcare in Uganda, where farmers depend on a diverse range of indigenous and introduced medicinal plants such as *Thymus vulgaris* for disease prevention and treatment. This community-rooted knowledge system offers an opportunity for sustainable, accessible, and culturally relevant livestock health management. However, its full potential remains underutilized due to limited phytochemical, pharmacological, and toxicological validation, as well as the absence of standardized dosages and safety profiles. To strengthen integration with conventional veterinary practice, future efforts should focus on scientific validation, policy recognition, and community engagement. National research institutions and universities should collaborate with traditional practitioners to conduct rigorous bioassays and establish dosage and safety standards. Policymakers, particularly within the Ministry of Agriculture, Animal Industry and Fisheries (MAAIF), should embed EVM into veterinary extension programs, regulatory frameworks, and One Health strategies. Meanwhile, documentation and preservation of orally transmitted knowledge should be prioritized through participatory community mapping and digital archiving. Such evidence-based and policy-supported integration will enhance animal health services, safeguard indigenous knowledge, and promote resilient, sustainable livestock systems across Uganda and beyond. Future research should prioritize systematic documentation, rigorous bioassays linked to specific diseases, and the establishment of dosage and safety standards. Such efforts will not only safeguard livestock health but also enhance rural livelihoods and preserve invaluable indigenous knowledge systems for generations to come.

CRediT authorship contribution statement

Godwin Anywar: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Arnold Ssegabo:** Methodology, Formal analysis, Data curation. **James Wanyama:** Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. **Caroline S. Weckerle:** Writing – review

& editing, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2025.120917>.

Data availability

No data was used for the research described in the article.

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